ENGR&204 — VISUAL GUIDE: match the picture, copy the template

How to use this on the exam (3 steps): 1. **Look** at your exam circuit → find the picture below that matches it. 2. **Copy** that template line-by-line, filling the blanks ____ with your numbers. 3. **Check** your answer's shape against the curve picture.

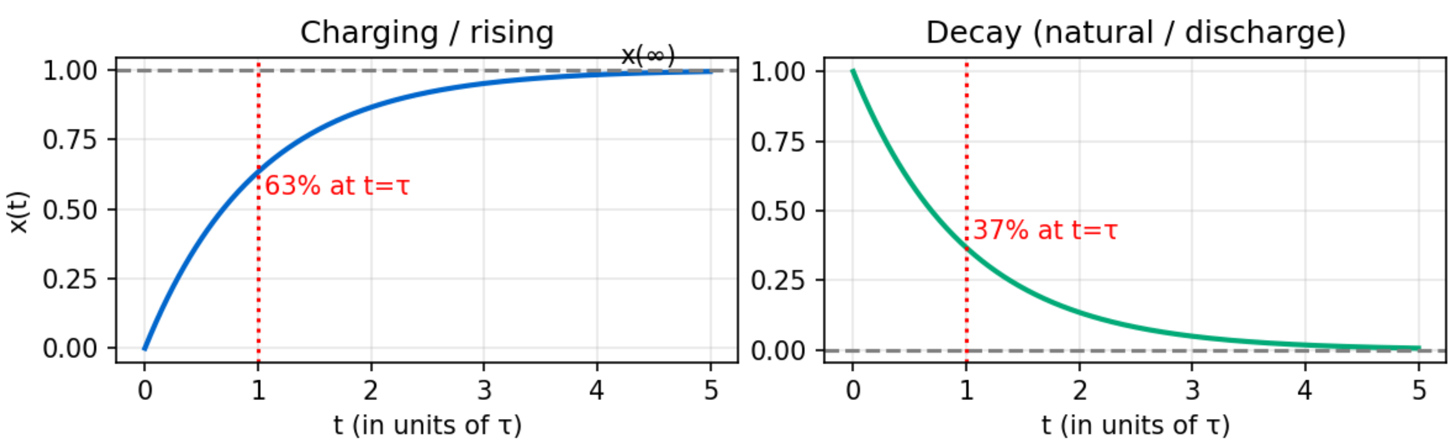
First, count the energy-storage elements (capacitors + inductors):

```
 0 caps/inductors  →  DC problem      →  use TYPE 5/6 (nodal, mesh, Thévenin, op-amp)
1 cap OR 1 inductor → FIRST-ORDER     →  use TYPE 1 (RC) or TYPE 2 (RL)  ← most of your exam
1 cap AND 1 inductor → SECOND-ORDER   →  use TYPE 3 (series RLC) or TYPE 4 (parallel RLC)
```

THE ONE FORMULA you use for every first-order problem

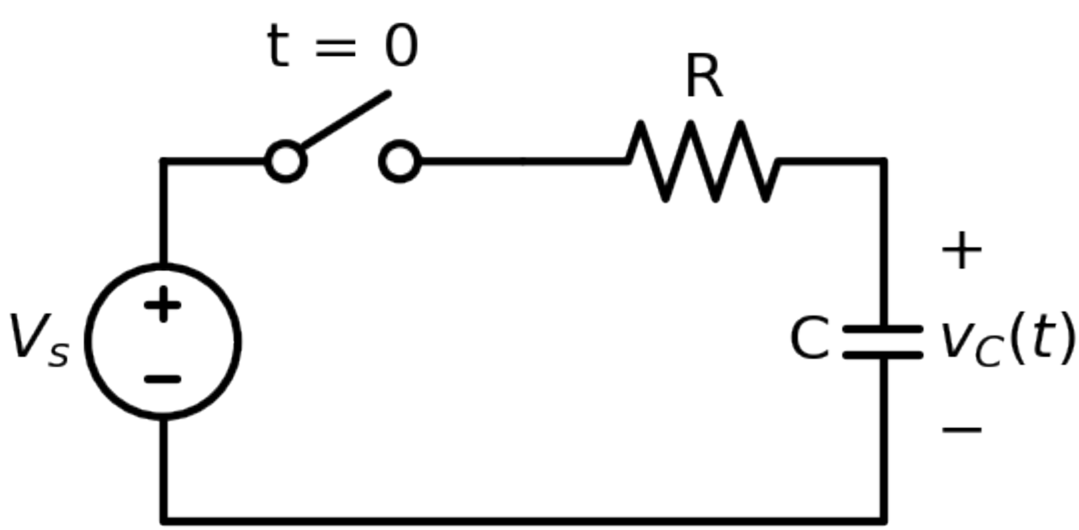
```
x(t) = x(∞) + [ x(0⁺) − x(∞) ] · e^( − t / τ )
       └ final ┘  └initial┘ └final┘         └ time constant ┘
```

`x` = the v(t) or i(t) you're asked for. You only ever need **3 numbers**: `x(0⁺)`, `x(∞)`, `τ`.



Left = answer rises to a final value (charging). Right = answer decays to 0 (natural/discharge). After 1τ you've moved 63% of the way; after 5τ it's basically done.

TYPE 1 — FIRST-ORDER RC (one capacitor)



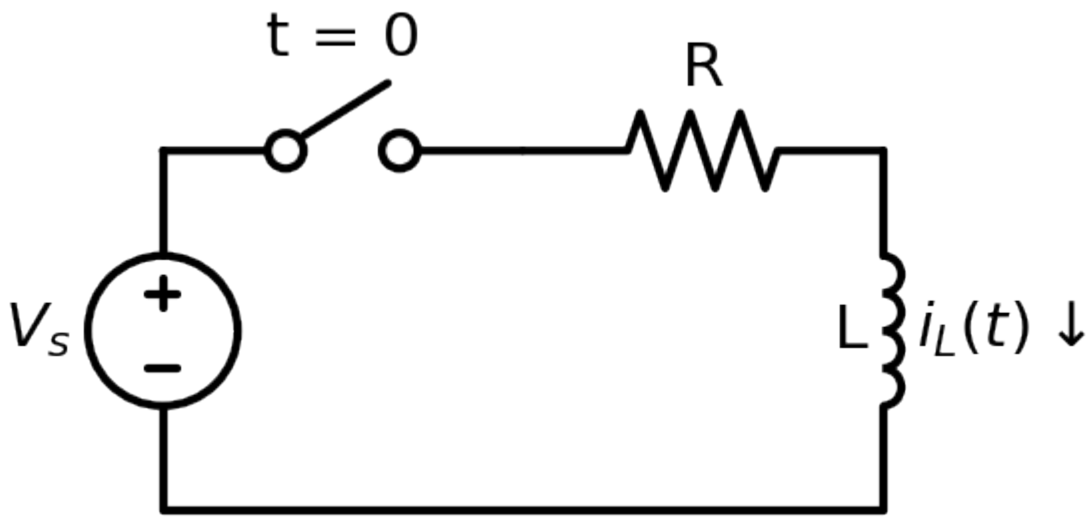
Recognize it: a **capacitor**, a switch, resistors, a DC source. Asked for `v_C(t)` or `v(t)`.

COPY THIS TEMPLATE (fill the blanks):

```
STEP 1  t<0 (before switch): capacitor = OPEN. Solve DC for the cap voltage.
        v_C(0⁻) = ______        →  v_C(0⁺) = v_C(0⁻) = ______   (cap voltage can't jump)

STEP 2  t→∞ (after switch settles): capacitor = OPEN. Solve DC again.
        v_C(∞) = ______

STEP 3  Time constant: turn OFF independent sources (V-source→wire, I-source→gap).
        Look into the capacitor's two terminals, find the resistance:  R_th = ______
        τ = R_th · C = ______ × ______ = ______ s

STEP 4  Plug in:
        v_C(t) = v_C(∞) + [ v_C(0⁺) − v_C(∞) ] e^(−t/τ)
        v_C(t) = ______ + ( ______ − ______ ) e^(−t/ ______ )   V
```

Worked (your HW6 #2): `v_C(0⁺)=4 V`, `v_C(∞)=6 V` ⇒ `v(t) = 6 + (4−6)e^(−t/τ) = 6 − 2e^(−t/τ) V`.

TYPE 2 — FIRST-ORDER RL (one inductor)



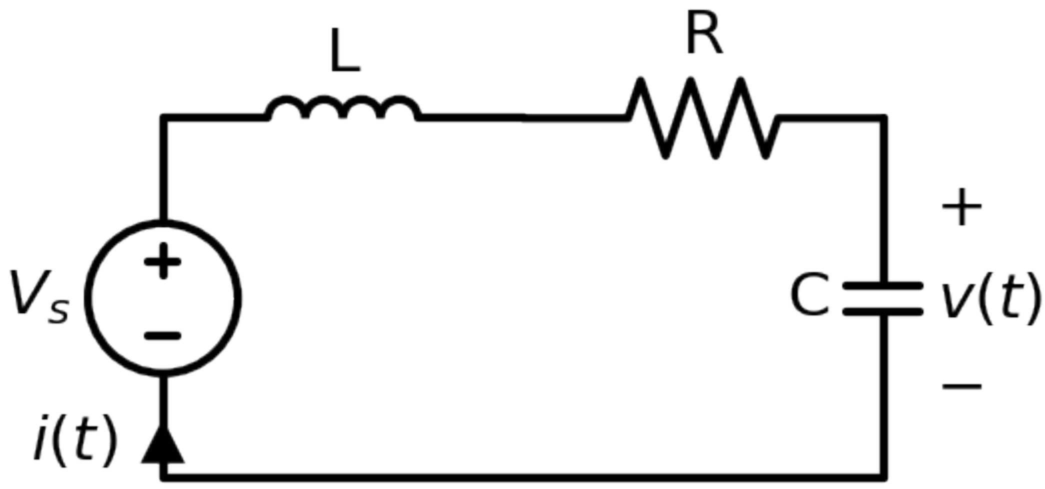
Recognize it: an **inductor**, a switch, resistors, a DC source. Asked for `i_L(t)` or `i(t)`.

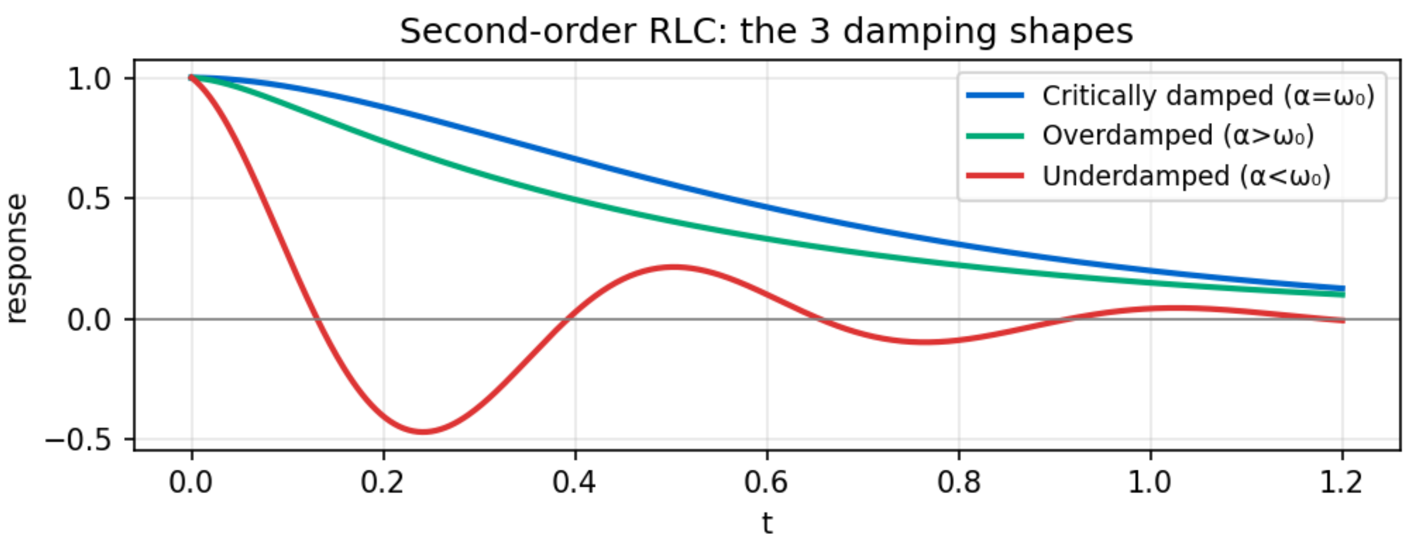
COPY THIS TEMPLATE:

```
STEP 1  t<0: inductor = SHORT (plain wire). Solve DC for the inductor current.
        i_L(0⁻) = ______       →  i_L(0⁺) = i_L(0⁻) = ______   (inductor current can't jump)

STEP 2  t→∞: inductor = SHORT. Solve DC again.
        i_L(∞) = ______

STEP 3  Time constant: turn OFF independent sources. Resistance seen by the inductor:
        R_th = ______          τ = L / R_th = ______ / ______ = ______ s

STEP 4  Plug in:
        i_L(t) = i_L(∞) + [ i_L(0⁺) − i_L(∞) ] e^(−t/τ)
        i_L(t) = ______ + ( ______ − ______ ) e^(−t/ ______ )   A
```

Need a voltage or another current? Get it from `v_L = L·di_L/dt`, or from Ohm's/KVL/KCL using `i_L(t)`. **Worked (Dorf P8.3-25):** read off the plot `v(0⁺)=100, v(∞)=20`, point (0.14s,60) ⇒ `τ=0.2s`, `v(t)=20+80e^(−5t) V`.

TYPE 3 — SECOND-ORDER SERIES RLC (R, L, C in one loop)





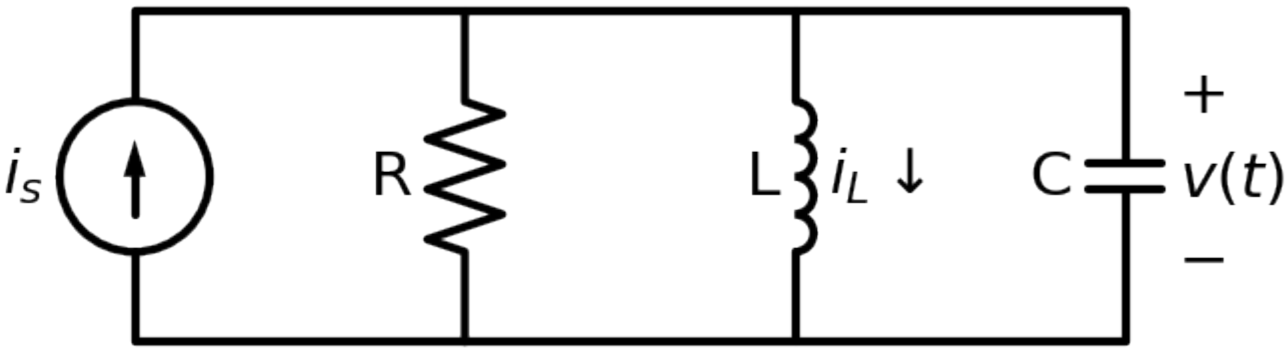
Recognize it: both an inductor and a capacitor **in a single loop** (after turning off sources). Usually solve for loop current `i(t)`.

COPY THIS TEMPLATE:

```
STEP 1  α = R / (2L) = ______              (SERIES: R over 2L)
STEP 2  ω₀ = 1/√(LC) = ______
STEP 3  Characteristic eqn:  s² + (R/L)s + 1/(LC) = 0   →   s² + ____ s + ____ = 0
        Roots:  s = −α ± √(α² − ω₀²) = ______ , ______
STEP 4  Pick the case by comparing α vs ω₀:
        α > ω₀  OVERDAMPED      → x(t) = A₁e^(s₁t) + A₂e^(s₂t)
        α = ω₀  CRITICAL        → x(t) = (A₁ + A₂t) e^(−αt)
        α < ω₀  UNDERDAMPED     → x(t) = e^(−αt)(B₁cos ω_d t + B₂sin ω_d t),  ω_d=√(ω₀²−α²)=____
STEP 5  Add forced part x(∞) if a source stays on;  find the 2 constants from:
        x(0⁺) = ______   and   x'(0⁺) = ______      (x'(0)= i_C(0)/C  or  v_L(0)/L)
```

Worked (handout P1): R=40, L=0.1, C=1/3 mF ⇒ $\alpha=200$, $\omega_0=173$ ⇒ $s^2+400s+30000=0$, roots **−100, −300** (overdamped).

TYPE 4 — SECOND-ORDER PARALLEL RLC (R, L, C all in parallel)



Recognize it: inductor and capacitor **side-by-side (parallel)** after turning off sources. Usually solve for node voltage `v(t)`.

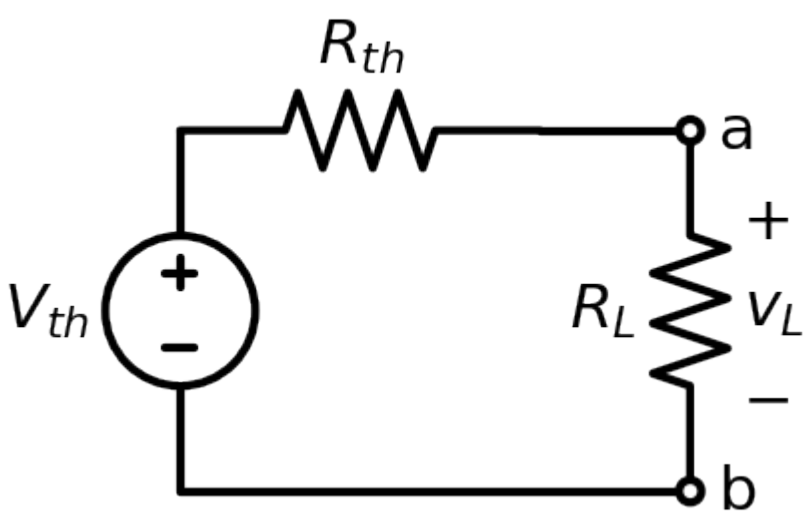
COPY THIS TEMPLATE (only Step 1 changes vs series!):

```
STEP 1  α = 1 / (2RC) = ______          (PARALLEL: 1 over 2RC)
STEP 2  ω₀ = 1/√(LC) = ______           (same as series)
STEP 3  Characteristic eqn:  s² + (1/RC)s + 1/(LC) = 0   →   s² + ____ s + ____ = 0
        Roots:  s = −α ± √(α² − ω₀²) = ______ , ______
STEP 4–5  same damping cases & constants as TYPE 3.
```

Worked (handout P3): R=10, L=½, C=5 mF ⇒ $\alpha=10$, $\omega_0=20$ ⇒ $s^2+20s+400=0$, roots **$-10 \pm j17.3$** (underdamped).

> **DAMPING TRAP:** in **series**, a large R is overdamped; in **parallel**, a small R is overdamped. (Series: $R>2\sqrt{L/C}$ over; Parallel: $R<\frac{1}{2}\sqrt{L/C}$ over.)

TYPE 5 — DC NETWORK: Thévenin / Norton / Max Power

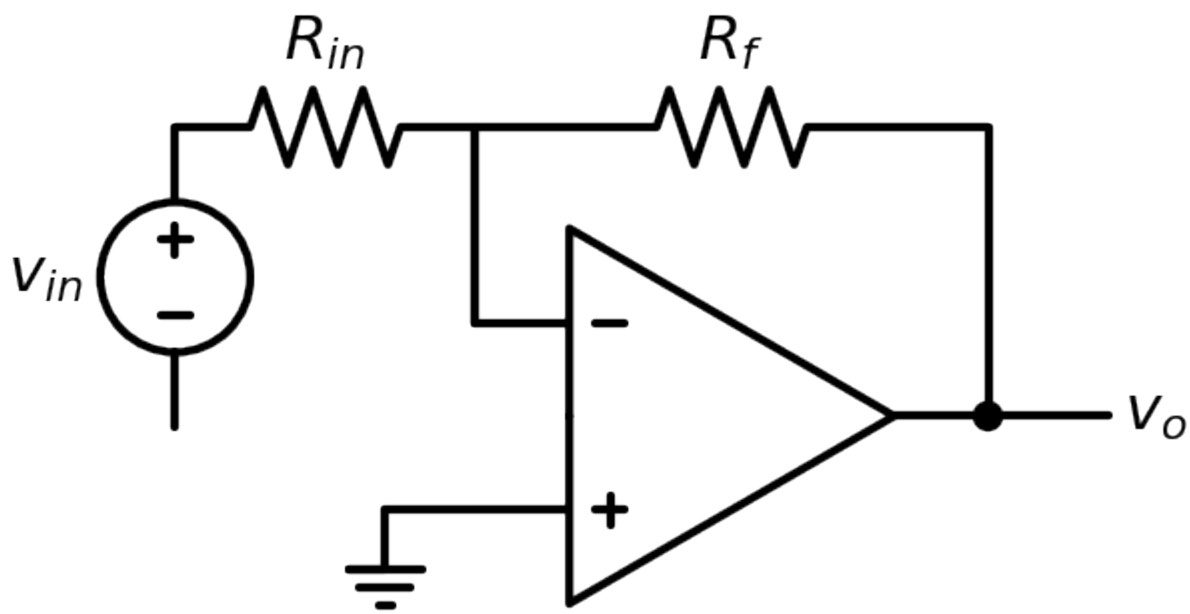


Recognize it: "find the equivalent seen by R_L", "max power", or a big resistor network. **Replace everything left of a–b with `V_th` in series with `R_th`.**

COPY THIS TEMPLATE:

```
V_th = open-circuit voltage across a–b (remove R_L, find V_oc) = ______
R_th = (turn off independent sources) resistance looking into a–b = ______
       [if dependent sources: R_th = V_oc / I_sc]
Current in a load R:   i = V_th / (R_th + R) = ______
MAX POWER:  happens when R_L = R_th ;   P_max = V_th² / (4 R_th) = ______
```

TYPE 6 — IDEAL OP-AMP

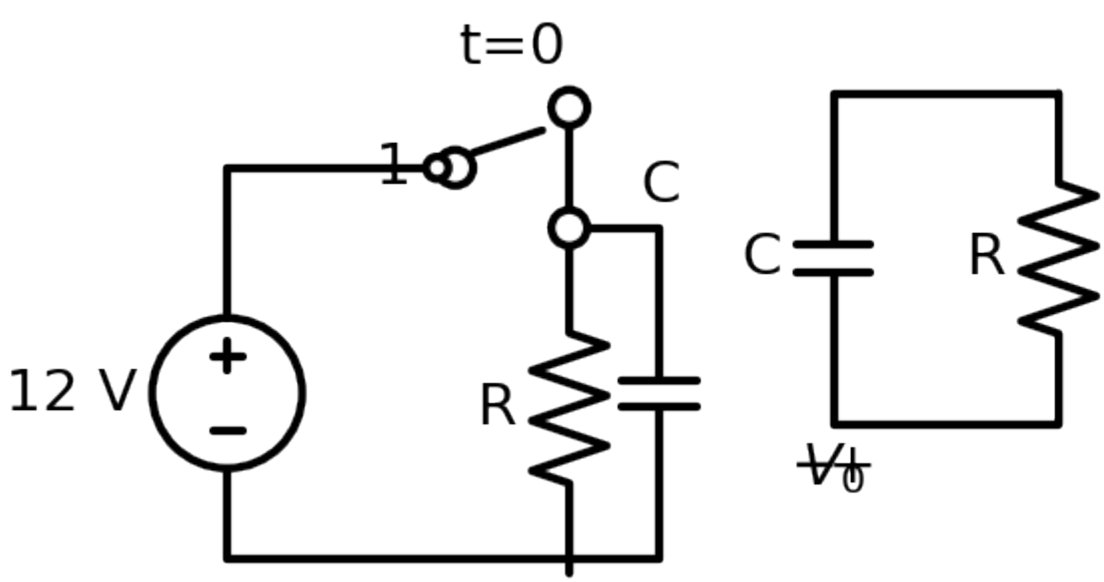


Recognize it: a triangle symbol. **Two golden rules:** (1) no current into the inputs $i_+ = i_- = 0$; (2) $v_+ = v_-$ (virtual short).
COPY THIS TEMPLATE:

```
 Find v₋ (= v₊, often = 0 "virtual ground" if + is grounded).
Write KCL at the inverting (−) node using i₊=i₋=0:
        (v_in − v₋)/R_in = (v₋ − v_o)/R_f
Solve for v_o.
Common results:  Inverting  v_o = −(R_f/R_in) v_in
                 Non-invert v_o = (1 + R_f/R_in) v_in
                 Output current i_o: KCL at the output node.
```

Worked (your HW2/HW5): 20kΩ in, 20kΩ → `V_M = −4 V`; `−2 − v_o = 8·3.5` → `v_o = −30 V`.

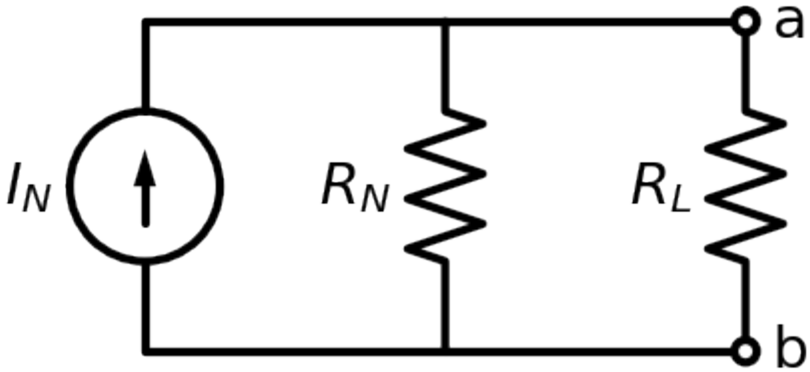
TYPE 7 — TWO-POSITION SWITCH / SOURCE-FREE (natural) transient



Recognize it: switch moves **1→2** at t=0, OR a charged C / energized L is left alone with just R (no source). This is the same first-order machine — the only change is `x(∞)`.
COPY THIS TEMPLATE:

```
Position 1 "for a long time" → find the stored state at the instant of switching:
        v_C(0) = ______  (cap charged through pos-1 path, DC)   /   i_L(0) = ______
After switch to position 2 (or source removed):
        If NO source remains:  x(∞) = 0   →   x(t) = x(0) e^(−t/τ)   (pure decay)
        τ = R_th·C  or  L/R_th using the position-2 resistances = ______
Energy released:  W_C = ½ C v_C(0)²   or   W_L = ½ L i_L(0)²  = ______ J
```

TYPE 8 — NORTON & SOURCE TRANSFORMATION

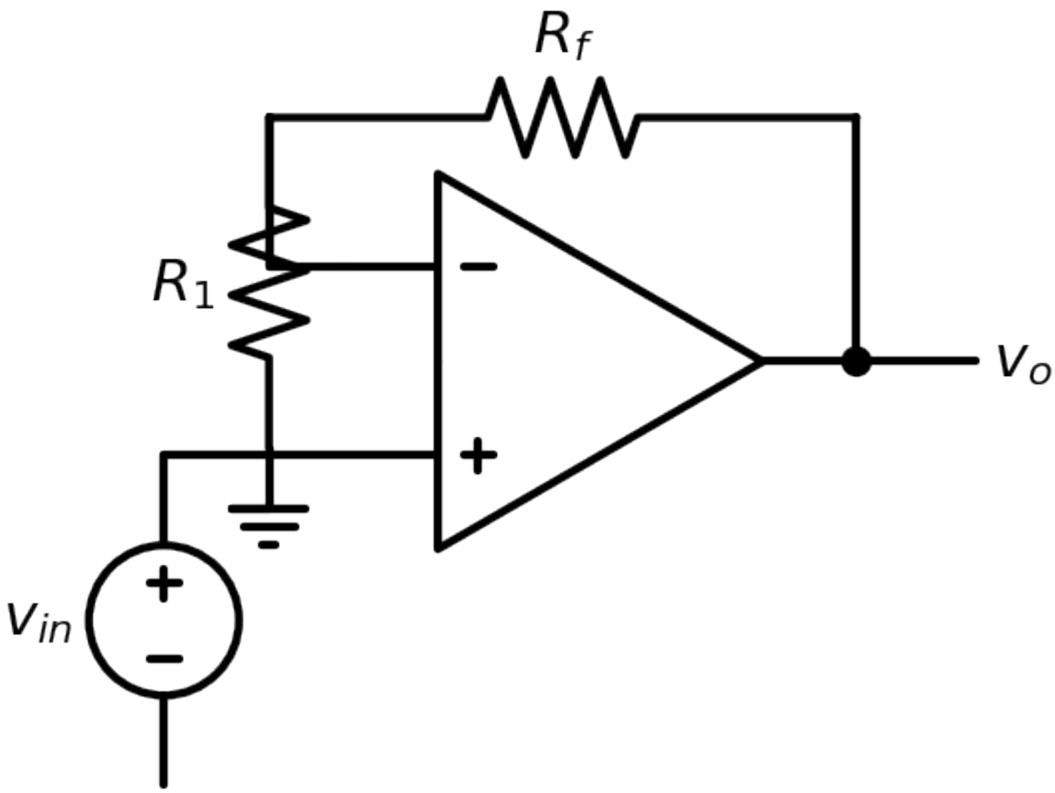


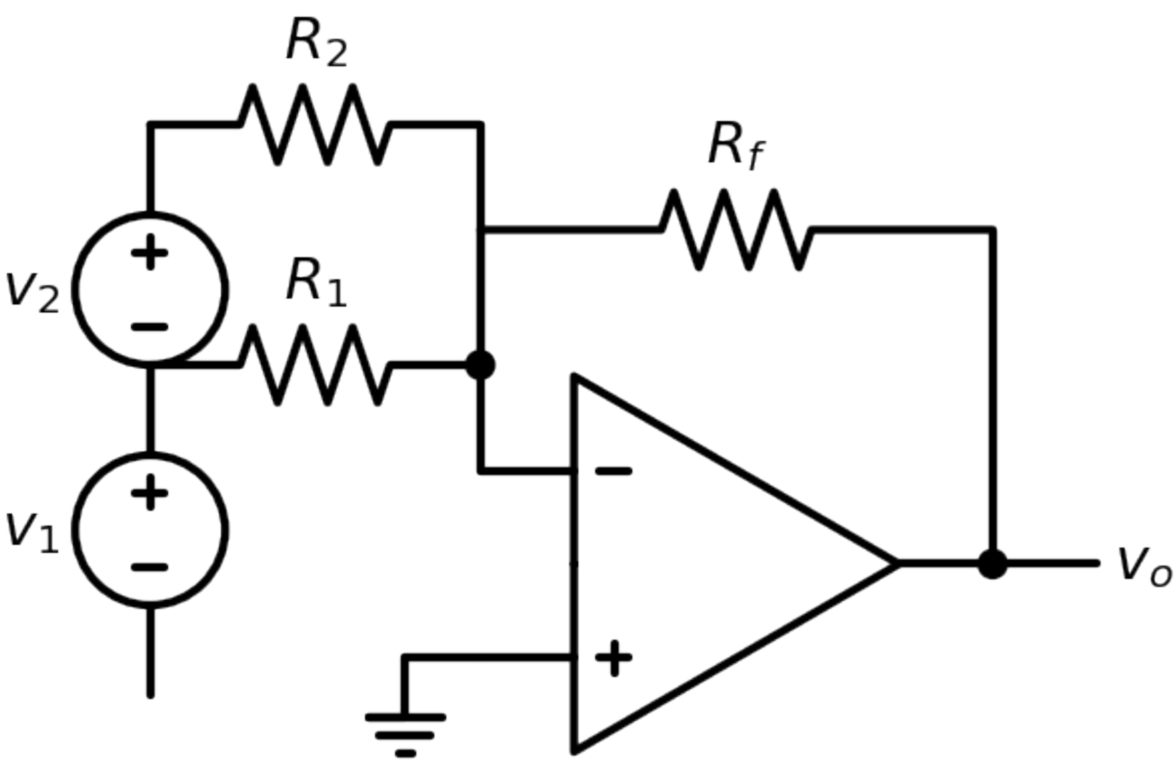
Recognize it: "Norton equivalent", or you want to simplify many sources/resistors.

COPY THIS TEMPLATE:

```
I_N = short-circuit current across a–b (put a wire on a–b, find i) = ______
R_N = R_th = (independent sources off) resistance into a–b = ______
Norton ⇄ Thévenin:   V_th = I_N · R_th ,   I_N = V_th / R_th ,   R_N = R_th
Source transform:  [V_s in series with R]  ⇄  [I_s = V_s/R in parallel with R]
```

TYPE 9 — OP-AMP VARIANTS

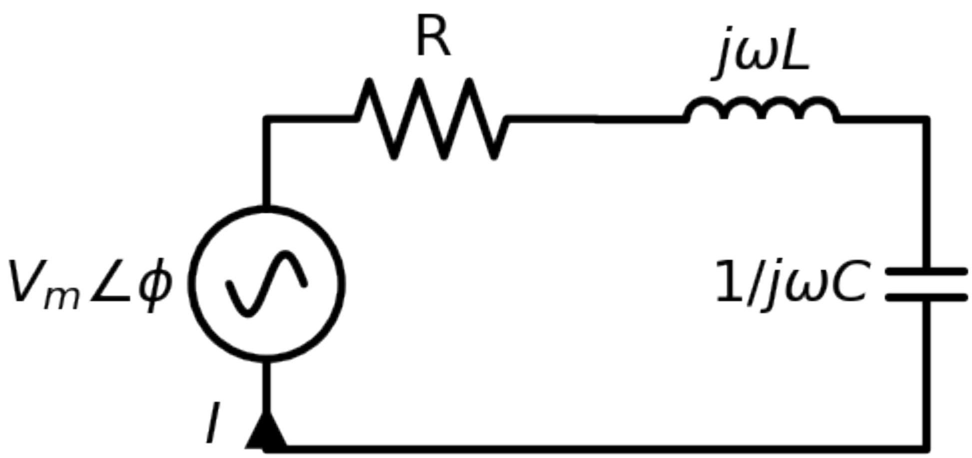




Golden rules: `i₊ = i₋ = 0` and `v₊ = v₋`. Then write KCL at the `−` node.

```
Inverting:      v_o = −(R_f/R_in) v_in
Non-inverting:  v_o = (1 + R_f/R_in) v_in
Buffer:         v_o = v_in
Summing:        v_o = −R_f ( v_1/R_1 + v_2/R_2 + … )
Difference:     v_o = (R_f/R_1)(v_2 − v_1)   [matched ratios]
Cascade:        multiply the stage gains
Output limited by supply rails:  V⁻ ≤ v_o ≤ V⁺  (saturation)
```

TYPE 10 — AC STEADY-STATE (phasors)

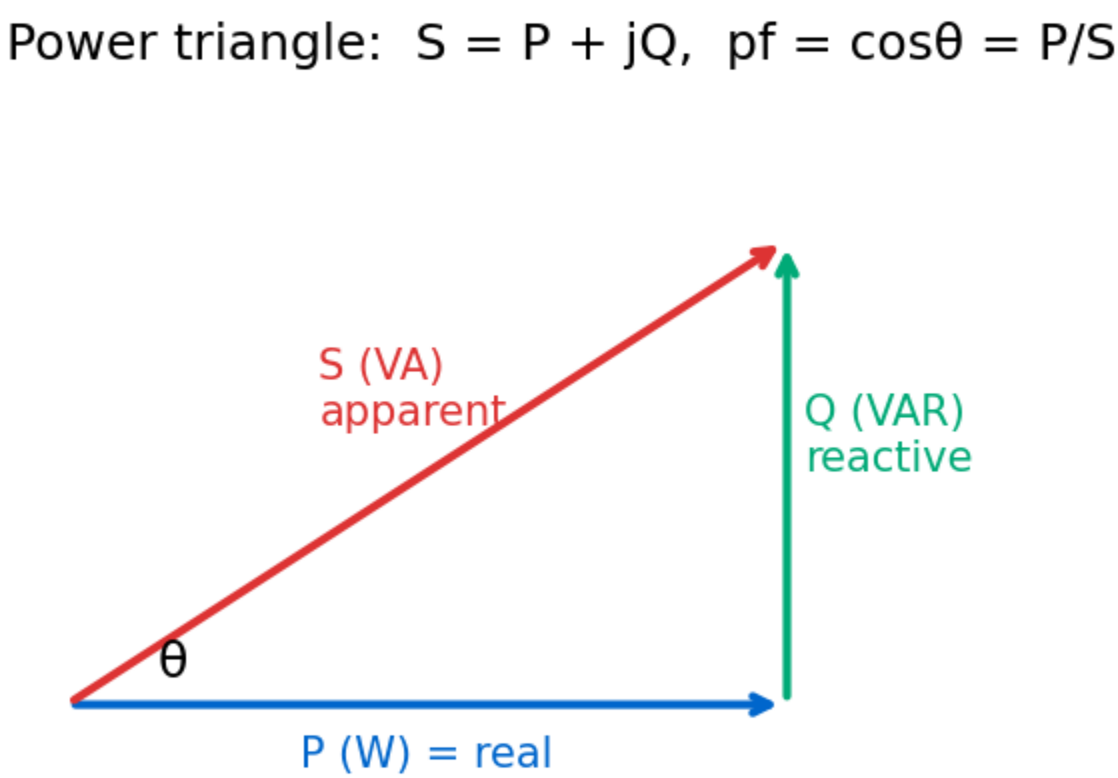


Recognize it: a sinusoidal source `V_m cos(ωt+φ)`; asked for steady-state v(t)/i(t).

COPY THIS TEMPLATE:

```
 1. Source → phasor:  V_m cos(ωt+φ) ⇒ V = V_m ∠φ
2. Elements → impedance at this ω:  Z_R=R,  Z_L=jωL,  Z_C=1/(jωC)=−j/(ωC)
3. Solve like DC but COMPLEX (Ohm V=IZ, dividers, nodal/mesh):
        I = V / Z_total = ______ ∠ ______
4. Back to time domain:  I_m∠θ ⇒ i(t)= I_m cos(ωt+θ)
Complex math:  add/sub in a+jb ; mult/div in r∠θ (mags ×/÷, angles +/−)
```

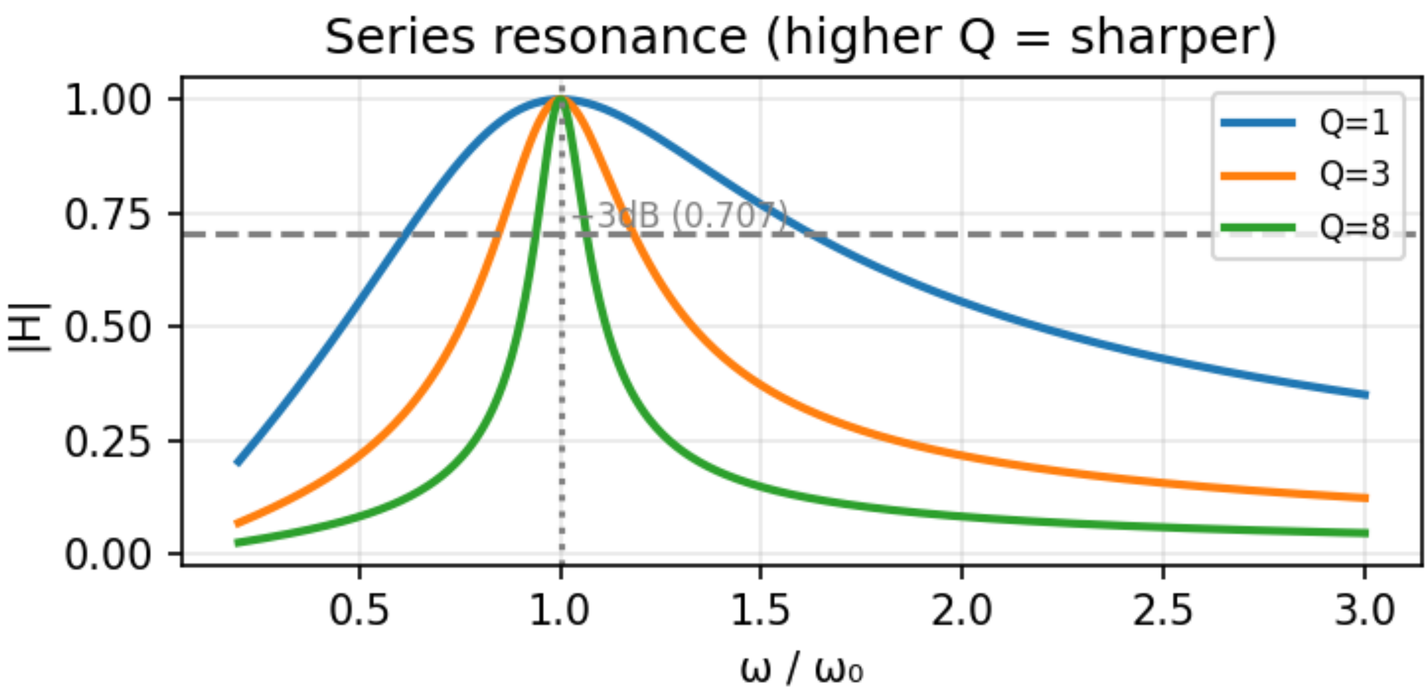
TYPE 11 — AC POWER & POWER-FACTOR CORRECTION

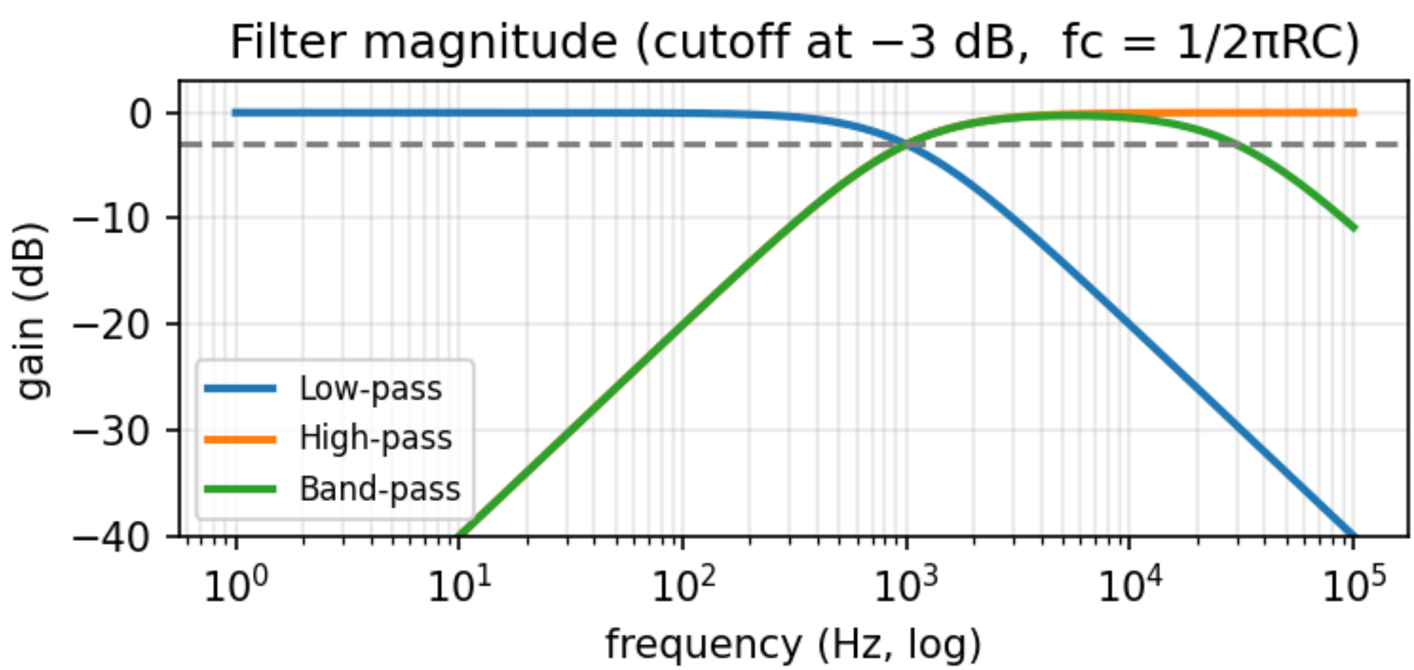


Use RMS values (`V_rms = V_m/√2`). With `θ = θ_v − θ_i`:

```
Real     P = V_rms I_rms cosθ   (W)       ← cosθ = power factor (pf)
React.  Q = V_rms I_rms sinθ   (VAR)     (+ inductive/lagging, − capacitive/leading)
Apparent S = V_rms I_rms        (VA)      Complex S = V·I* = P + jQ ;  |S|=√(P²+Q²)
Per element:  P = I_rms²R ,  Q = I_rms²X
PF correction (add parallel C):  Q_C = P(tanθ_old − tanθ_new),  C = Q_C/(ω V_rms²)
```

TYPE 12 — RESONANCE & FILTERS



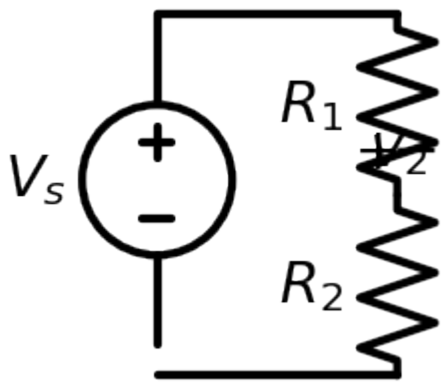


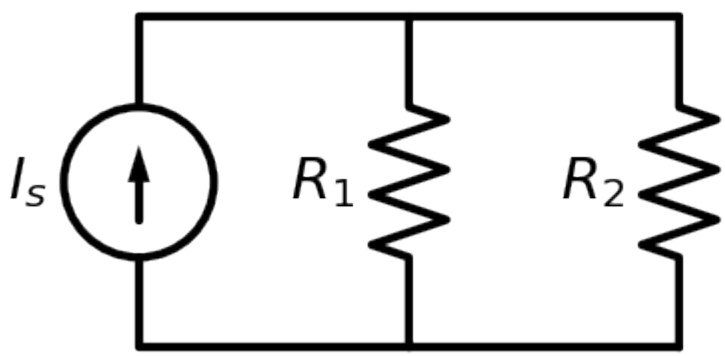
```
Resonance (series RLC): ω₀ = 1/√(LC) ; at ω₀, Z is purely real (=R), current is MAX.
Quality factor:  Q = ω₀L/R = 1/(ω₀RC)        Bandwidth: BW = ω₀/Q
RC filter cutoff:  f_c = 1/(2πRC)   (gain is 0.707 / −3 dB at f_c)
Low-pass: output across C.  High-pass: output across R.  Band-pass: cascade LP+HP.
Band-pass center (geometric mean): f_r = √(f_L · f_H)
```

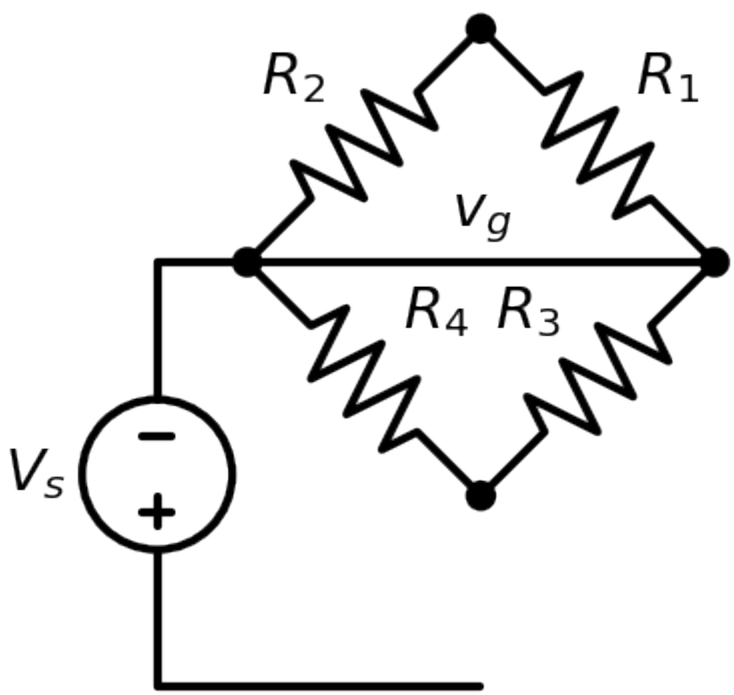
TYPE 13 — NODAL / MESH (when there's no single divider trick)

```
NODAL (pick when fewer nodes):                      MESH (pick when fewer loops):
1. Ground the bottom node.                          1. Clockwise mesh currents i₁,i₂…
2. Node voltages V₁,V₂…                              2. KVL each mesh; shared R carries (i_this−i_adj)
3. KCL each node: Σ (V_this−V_other)/R = sources 3. Supermesh if a current source is shared
4. Supernode if a V-source links two nodes          4. Solve the linear system
5. Solve the linear system
```

TYPE 14 — DIVIDERS, Δ-Y, & WHEATSTONE BRIDGE








```
Voltage divider (series): v_k = V_s · R_k / ΣR
Current divider (2 parallel): i_1 = i_total · R_2/(R_1+R_2)   (current favors SMALLER R)
Δ→Y:  R_Y = (product of the two adjacent Δ resistors) / (sum of all three Δ resistors)
Y→Δ:  R_Δ = (sum of pairwise products of Y resistors) / (the opposite Y resistor)
Balanced Wheatstone bridge (v_g = 0):  R_1/R_2 = R_3/R_4   (use to find an unknown R)
```

TYPE 15 — CHARGE & ENERGY FROM A GRAPH

```
Charge transferred:  q = ∫ i dt  = AREA under the i–t graph
Energy:              W = ∫ p dt = ∫ v·i dt
Method: break the graph into straight-line pieces; write v(t),i(t) for each interval;
        multiply p=v·i; integrate each interval; add them up.
```

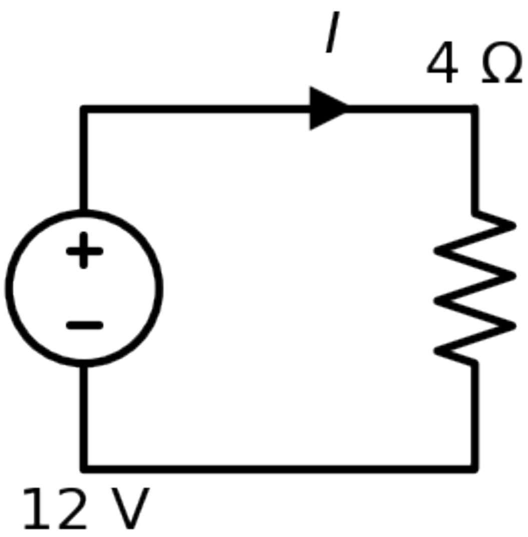
Diagrams generated for this guide; component values are placeholders — drop in your exam's numbers. Full algebra for each worked example is in *`WORKED_EXAMPLES.md`*; all formulas in *`CHEATSHEET.md`*.

ENGR&204 — SOLVED PROBLEMS BANK (diagram + walkthrough for every type)

Each problem below has a **circuit diagram with real numbers** and an **easy-to-read step-by-step solution**. Find the one that looks like your exam problem and follow the same steps with your numbers. Ordered **basic → complex**.

BASIC

S1 — Ohm's Law & Power

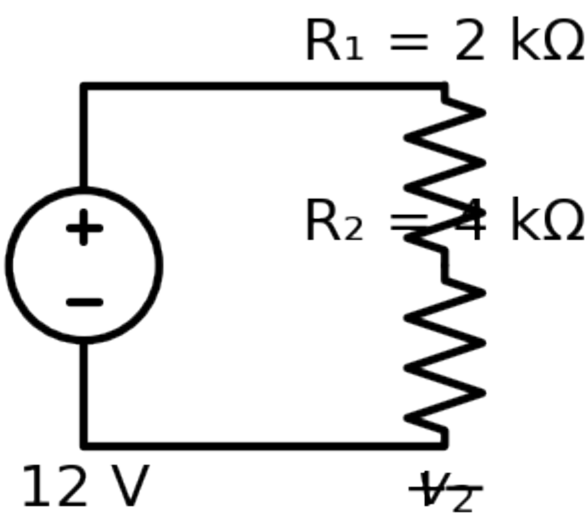


Given: 12 V across a 4 Ω resistor. Find I and power.

```
 I = V/R = 12/4 = 3 A
P = V·I = 12·3 = 36 W        (check: I²R = 3²·4 = 36 W ✓)
```

Answer: I = 3 A, P = 36 W.

S2 — Series resistors & Voltage Divider

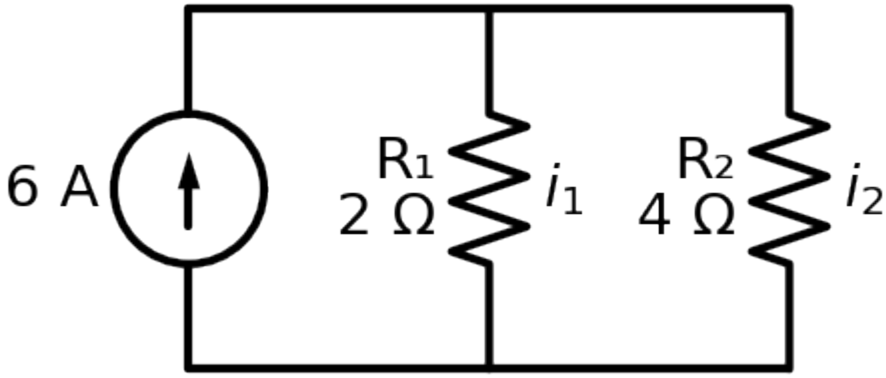


Given: 12 V, R_1 = 2 kΩ in series with R_2 = 4 kΩ. Find v_2.

```
 Same current: I = V/(R₁+R₂) = 12/(6 kΩ) = 2 mA
Divider:  v₂ = V·R₂/(R₁+R₂) = 12·4/6 = 8 V     (v₁ = 4 V)
```

Answer: v_2 = 8 V.

S3 — Parallel resistors & Current Divider

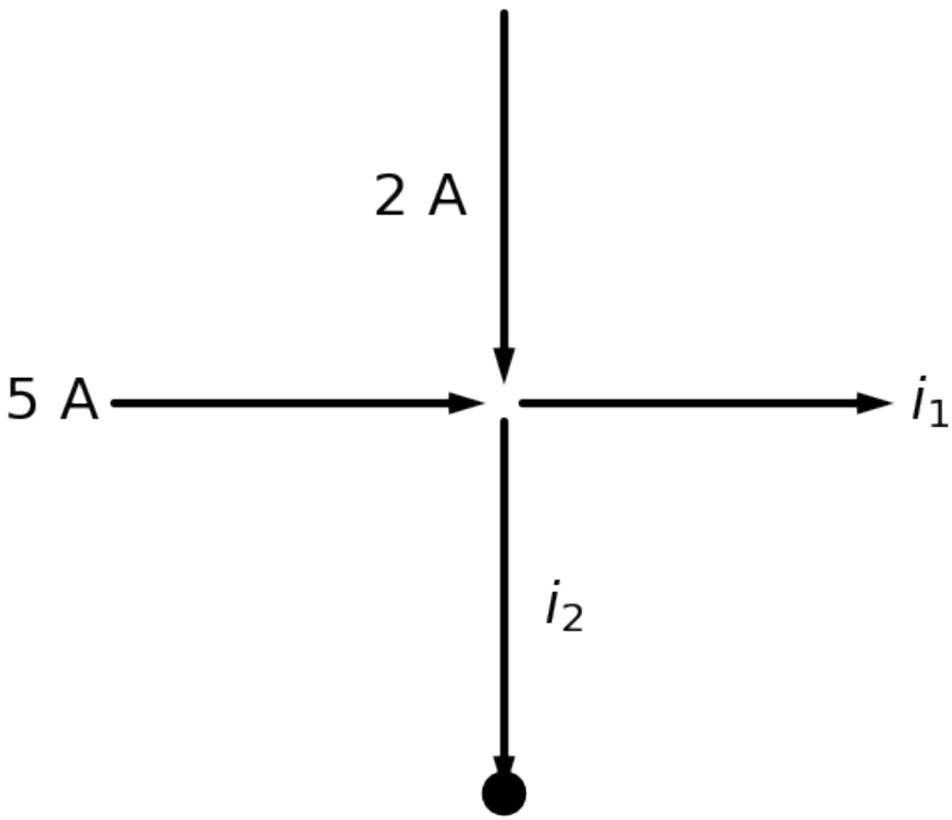


Given: 6 A into $R_1 = 2\ \Omega \parallel R_2 = 4\ \Omega$. Find i_1, i_2.

```
Current divider (current favors the SMALLER R):
   i₁ = 6·R₂/(R₁+R₂) = 6·4/6 = 4 A
   i₂ = 6·R₁/(R₁+R₂) = 6·2/6 = 2 A          (check 4+2 = 6 ✓)
```

Answer: $i_1 = 4$ A, $i_2 = 2$ A.

S4 — KCL at a node

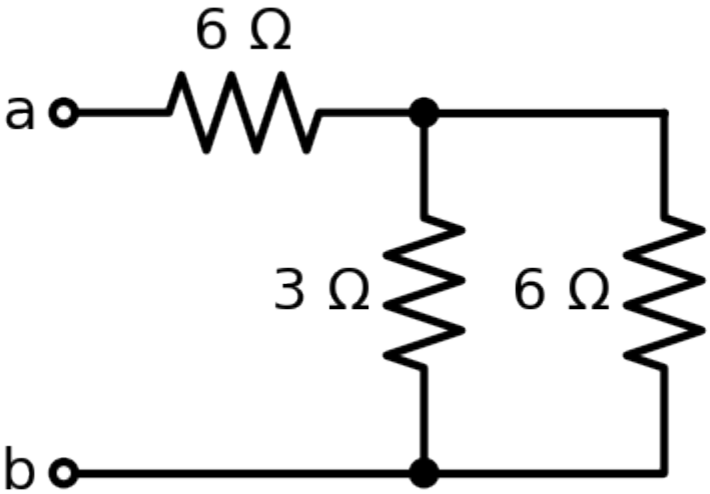


Given: 5 A and 2 A flow in; i_1 and i_2 flow out; i_1 measured = 3 A. Find i_2.

```
KCL:  Σ in = Σ out  →  5 + 2 = i₁ + i₂  →  7 = 3 + i₂  →  i₂ = 4 A
```

Answer: $i_2 = 4$ A.

S5 — Equivalent Resistance (series–parallel reduction)

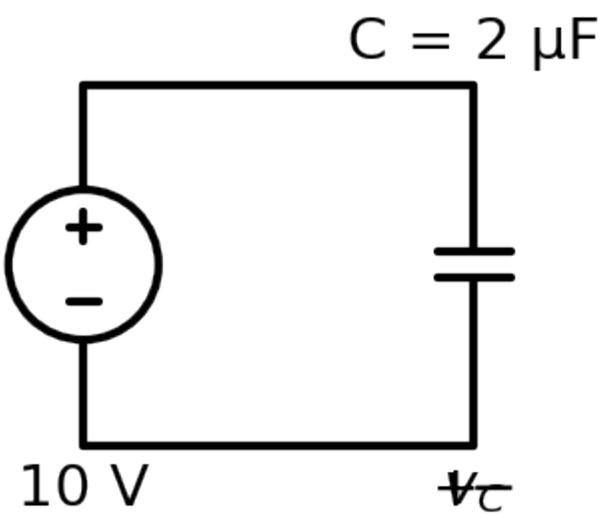


Given: 6 Ω in series with (3 Ω ‖ 6 Ω). Find R_ab.

```
Parallel first:  3‖6 = (3·6)/(3+6) = 2 Ω
Then series:     R_ab = 6 + 2 = 8 Ω
```

Answer: R_ab = 8 Ω.

S6 — Capacitor stored charge & energy



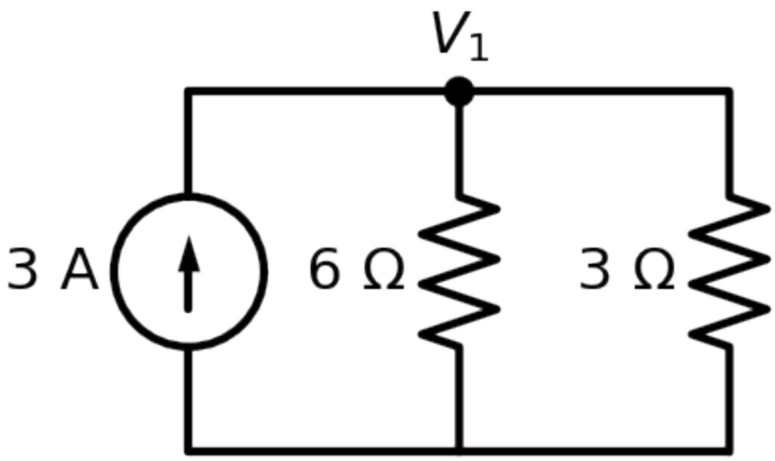
Given: C = 2 µF charged to 10 V.

```
Charge:  q = C·V = 2µ·10 = 20 µC
Energy:  W = ½C·V² = ½·2µ·10² = 100 µJ
(DC steady state: a capacitor is an OPEN circuit — no current through it.)
```

Answer: q = 20 µC, W = 100 µJ.

INTERMEDIATE (DC network analysis)

S7 — Nodal Analysis

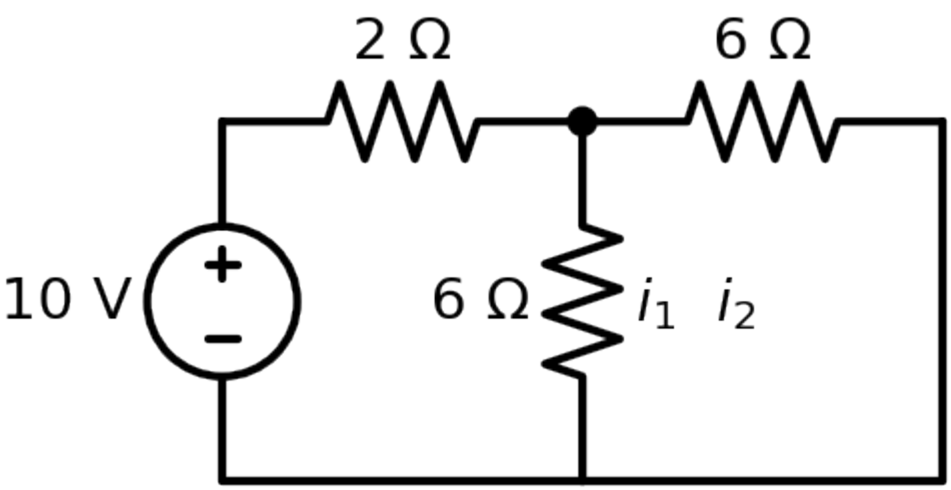


Given: 3 A source feeding node V_1 with 6 Ω and 3 Ω to ground. Find V_1.

```
KCL at V₁ (currents leaving = source in):
   V₁/6 + V₁/3 = 3   →   V₁(1/6 + 1/3) = 3   →   V₁·(1/2) = 3   →   V₁ = 6 V
Branch currents: 6Ω → 1 A, 3Ω → 2 A  (sum = 3 A ✓)
```

Answer: $V_1 = 6$ V.

S8 — Mesh Analysis

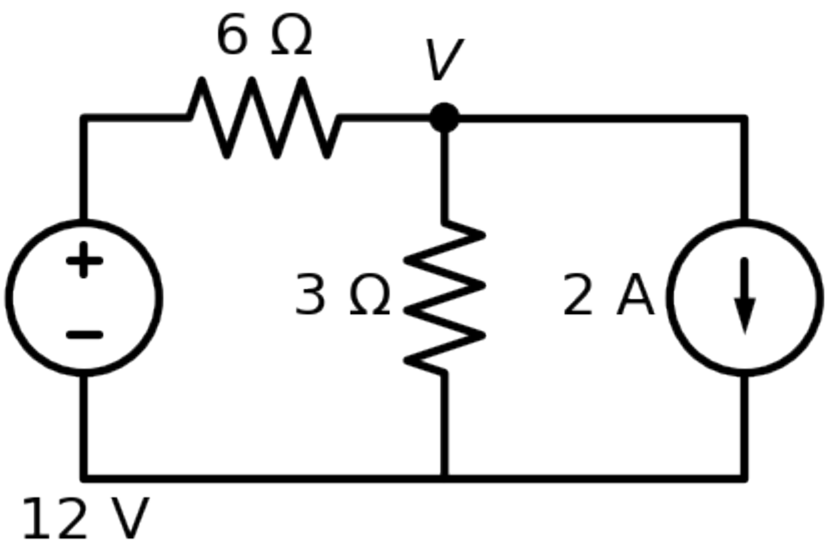


Given: 10 V, $R_1 = 2$ Ω (mesh 1), shared 6 Ω, $R_3 = 6$ Ω (mesh 2). Find mesh currents.

```
Mesh 1:  −10 + 2·i₁ + 6(i₁ − i₂) = 0   →   8i₁ − 6i₂ = 10
Mesh 2:  6(i₂ − i₁) + 6·i₂ = 0          →   −6i₁ + 12i₂ = 0  →  i₂ = ½ i₁
Substitute:  8i₁ − 3i₁ = 10  →  5i₁ = 10  →  i₁ = 2 A,  i₂ = 1 A
```

Answer: $i_1 = 2$ A, $i_2 = 1$ A.

S9 — Superposition

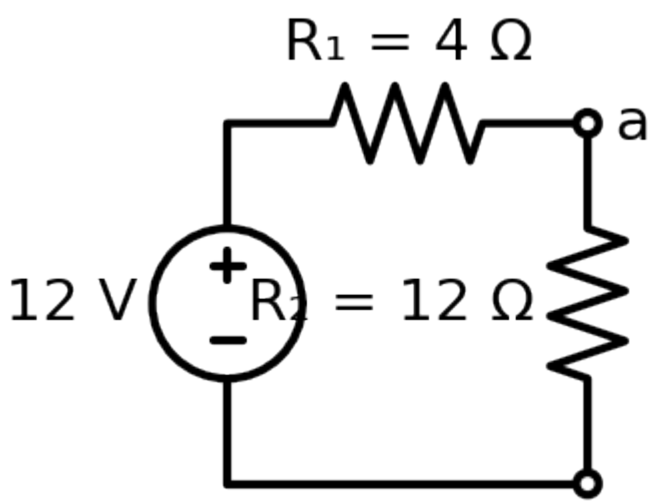


Given: 12 V (via 6 Ω) and a 2 A source share node V across 3 Ω. Find V.

```
ONE source at a time:
• 12 V only (open the 2 A source):  V_a = 12·3/(6+3) = 4 V
• 2 A only (short the 12 V source): 6‖3 = 2 Ω,  V_b = 2·2 = 4 V
Add:  V = V_a + V_b = 4 + 4 = 8 V       (check by nodal: (V−12)/6 + V/3 = 2 ✓)
```

Answer: V = 8 V.

S10 — Thévenin Equivalent

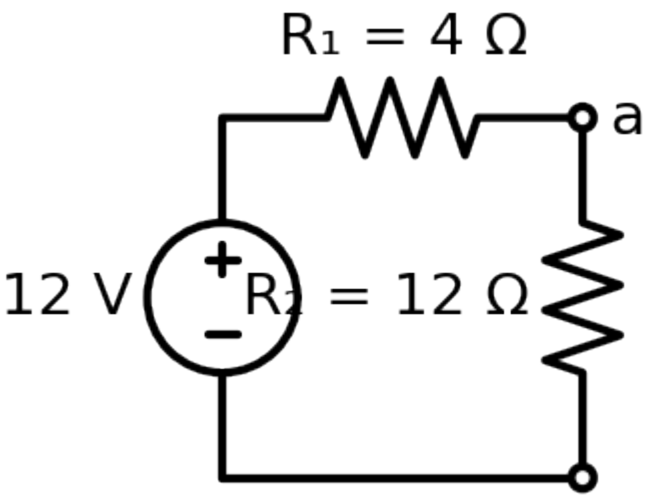


Given: 12 V, $R_1 = 4\ \Omega$, internal $R_2 = 12\ \Omega$. Find Thévenin at a–b.

```
V_th = V_oc = voltage across R₂ = 12·12/(4+12) = 9 V
R_th = (short the 12 V source) R₁ ‖ R₂ = 4‖12 = 3 Ω
```

Answer: V_th = 9 V, R_th = 3 Ω.

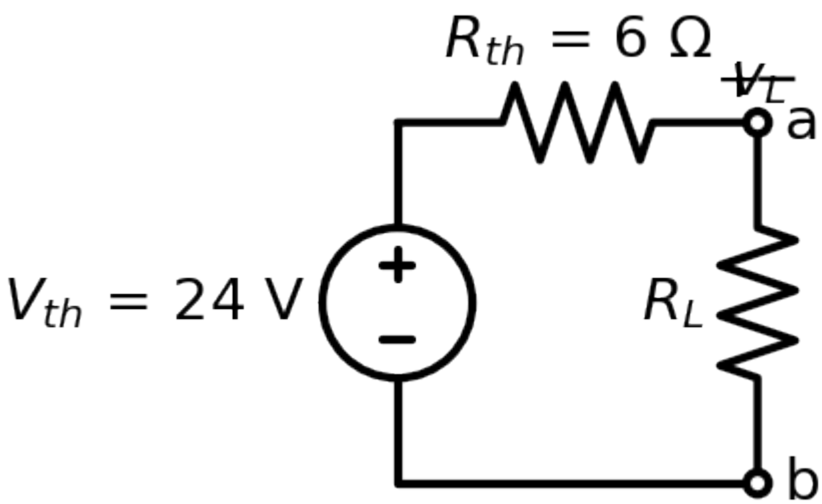
S11 — Norton Equivalent (same circuit as S10)




```
I_N = I_sc (short a–b): the 12 Ω is shorted, so I_sc = 12 V / 4 Ω = 3 A
R_N = R_th = 3 Ω
Check: I_N = V_th/R_th = 9/3 = 3 A ✓   (Norton ⇄ Thévenin)
```

Answer: I_N = 3 A, R_N = 3 Ω.

S12 — Maximum Power Transfer

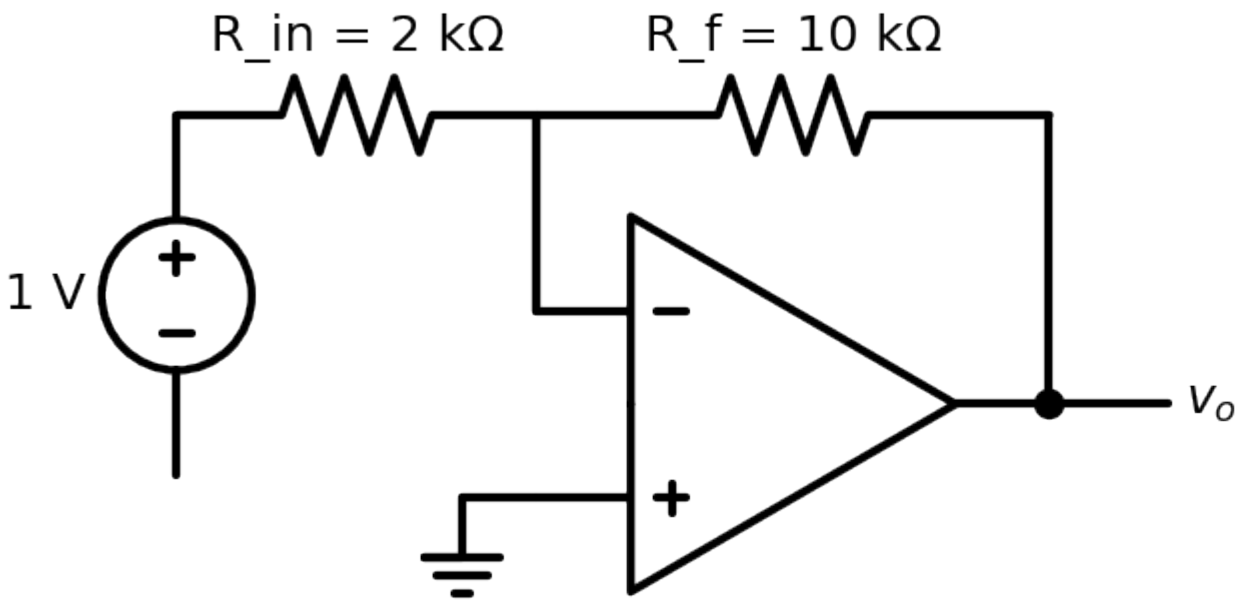


Given: V_th = 24 V, R_th = 6 Ω. Find R_L and P_max.

```
Max power when  R_L = R_th = 6 Ω
P_max = V_th²/(4·R_th) = 24²/(4·6) = 576/24 = 24 W
```

Answer: R_L = 6 Ω, P_max = 24 W.

S13 — Inverting Op-Amp

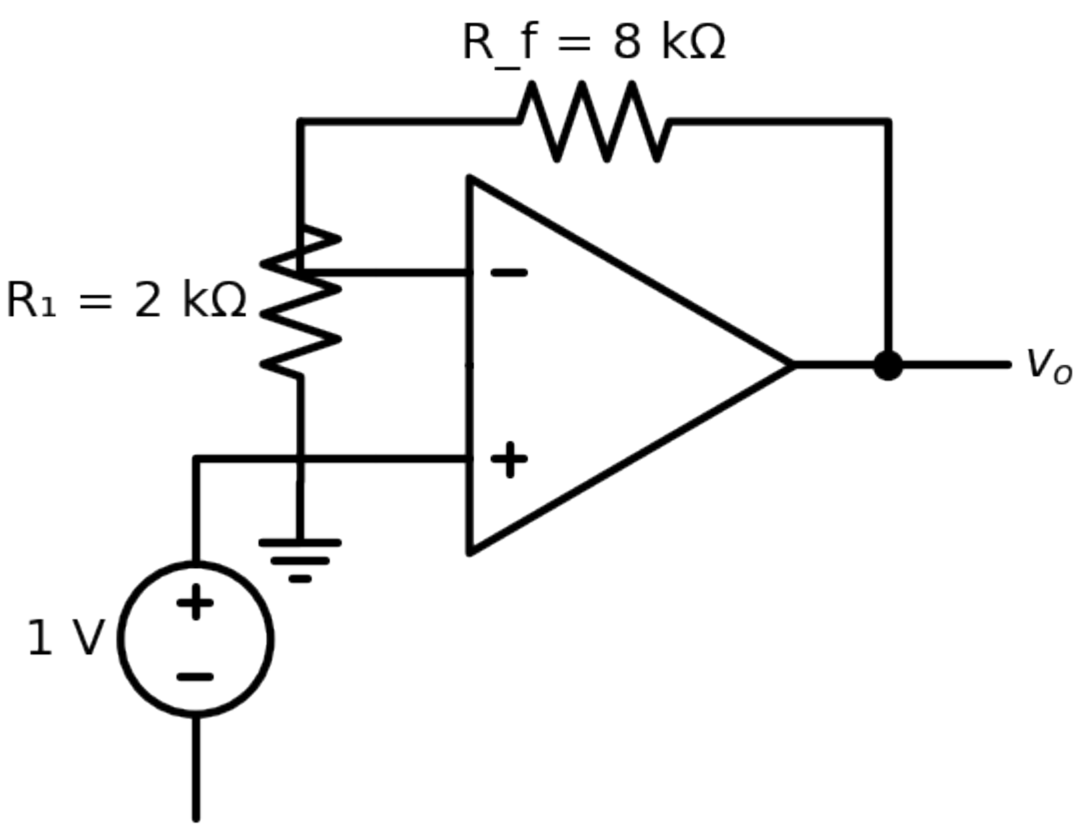


Given: R_in = 2 kΩ, R_f = 10 kΩ, v_in = 1 V.

```
v_o = −(R_f/R_in)·v_in = −(10/2)·1 = −5 V
```

Answer: v_o = −5 V.

S14 — Non-Inverting Op-Amp



Given: R_1 = 2 kΩ, R_f = 8 kΩ, v_in = 1 V.

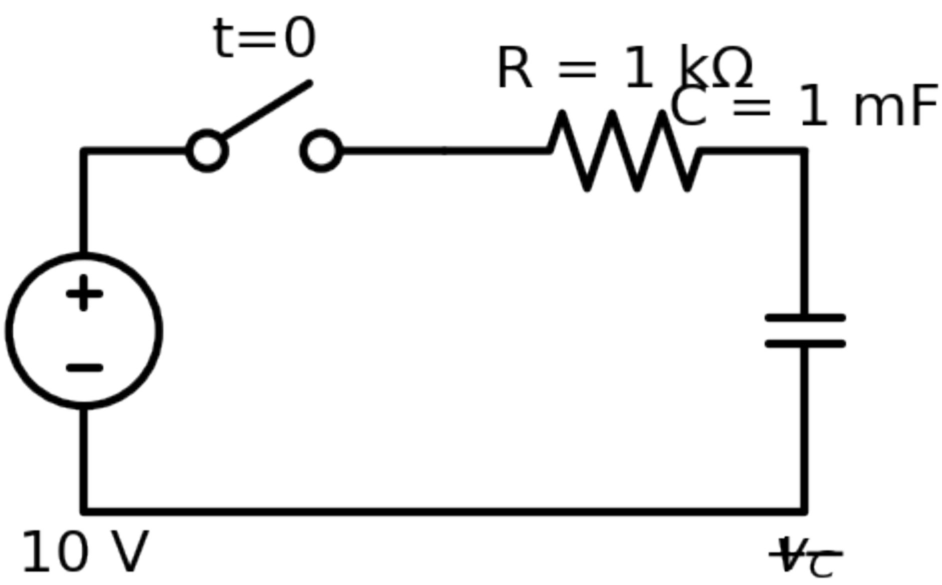
```
v_o = (1 + R_f/R₁)·v_in = (1 + 8/2)·1 = 5 V
```

Answer: v_o = 5 V.

░ FIRST-ORDER TRANSIENTS (core of the final) ░

> Every one uses: **x(t) = x(∞) + [x(0⁺) − x(∞)]·e^(−t/τ)**.

S15 — RC Charging

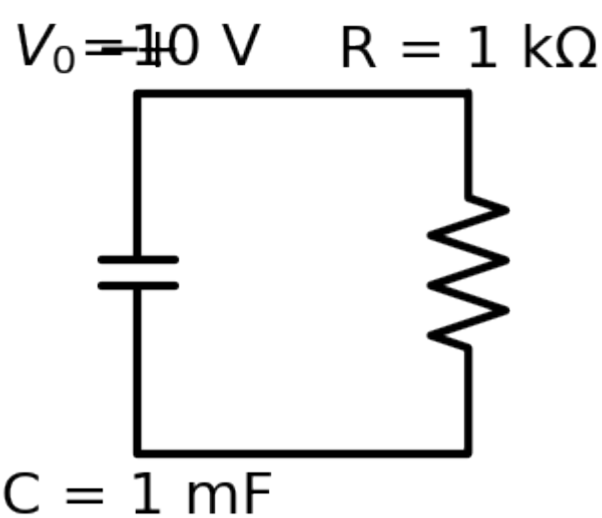


Given: 10 V, switch closes at t=0, R = 1 kΩ, C = 1 mF. Find v_C(t).

```
x(0⁺): cap uncharged → v_C(0⁺) = 0 V
x(∞):  cap = open at DC → v_C(∞) = 10 V
τ = R·C = 1000·0.001 = 1 s
v_C(t) = 10 + (0 − 10)e^(−t/1) = 10(1 − e^(−t)) V
i(t)   = (10/1k)e^(−t) = 10 e^(−t) mA
```

Answer: v_C(t) = 10(1 − e^(−t)) V.

S16 — RC Discharging (source-free)

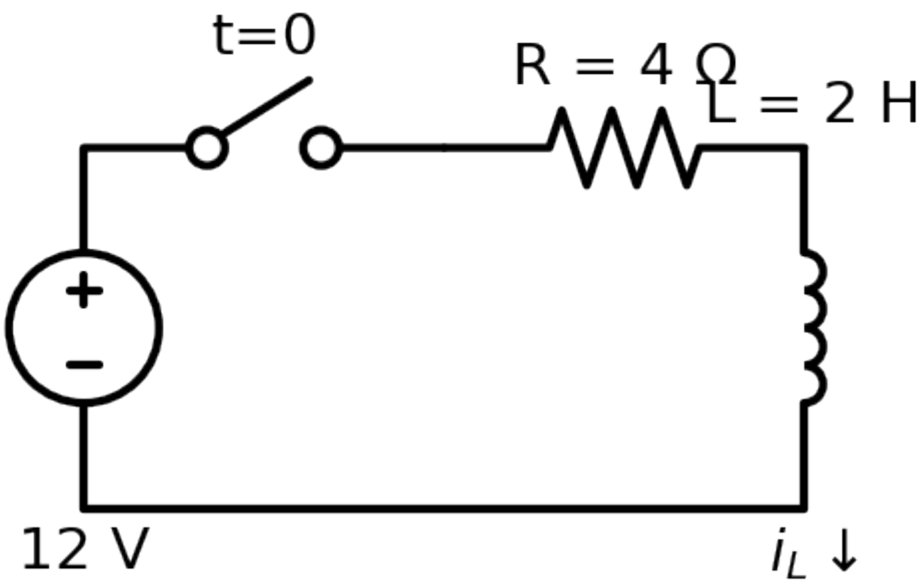


Given: C = 1 mF charged to V_0 = 10 V, discharges through R = 1 kΩ. Find v_C(t).

```
x(0⁺) = 10 V,  x(∞) = 0 (no source),  τ = RC = 1 s
v_C(t) = 10 e^(−t) V        (pure decay)
```

Answer: v_C(t) = 10 e^(−t) V.

S17 — RL Step Response



Given: 12 V, switch at t=0, R = 4 Ω, L = 2 H. Find i_L(t).

```
x(0⁺): inductor carried 0 before → i_L(0⁺) = 0 A
x(∞):  inductor = short at DC → i_L(∞) = 12/4 = 3 A
τ = L/R = 2/4 = 0.5 s    (so 1/τ = 2)
i_L(t) = 3 + (0 − 3)e^(−t/0.5) = 3(1 − e^(−2t)) A
v_L(t) = L·di/dt = 12 e^(−2t) V
```

Answer: i_L(t) = 3(1 − e^(−2t)) A.

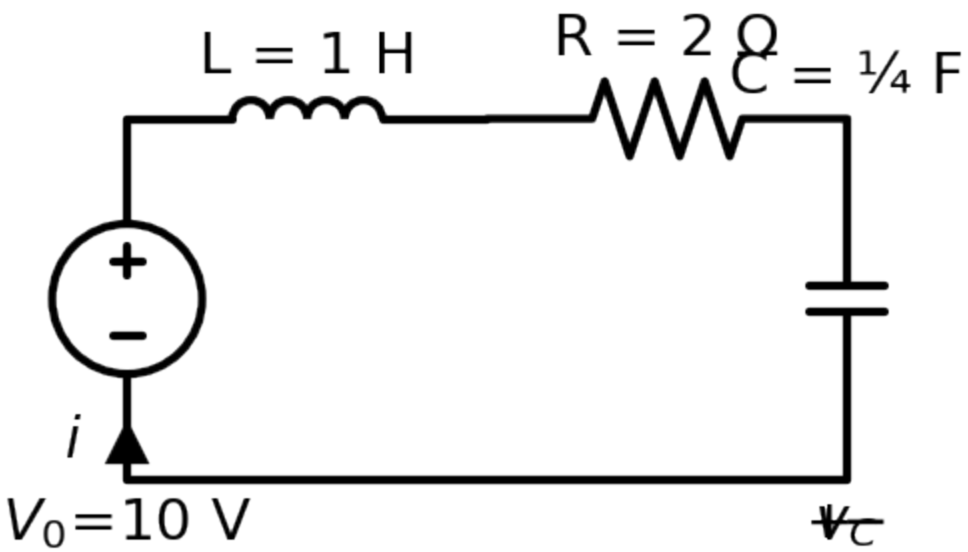
S18 — First-order with a DEPENDENT source

(diagram & full work: *WORKED_EXAMPLES.md* A3 — Dorf P8.7-1)

```
Method: can't use τ=RC blindly. Derive the ODE  dv/dt + a·v = forcing,
then v = (natural e^(−at)) + (forced, matching the source).
Result:  v_C(t) = 4 e^(−9t) + 18 e^(−5t) V.
```

SECOND-ORDER RLC (heavily tested)

S19 — Series RLC: characteristic equation, roots, complete response

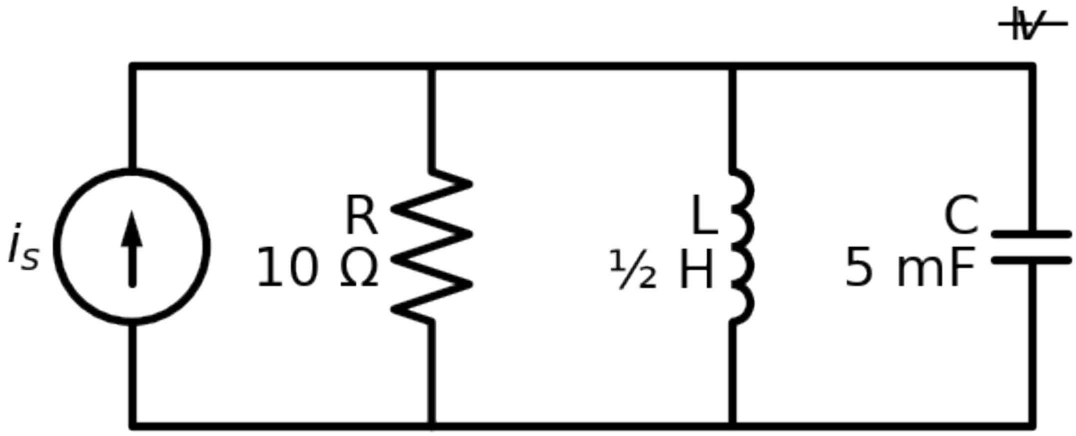


Given: source-free series, R = 2 Ω, L = 1 H, C = ¼ F, with i(0) = 0 and v_C(0) = 10 V.

```
α = R/(2L) = 2/2 = 1
ω₀ = 1/√(LC) = 1/√(1·0.25) = 2
α < ω₀  ⇒ UNDERDAMPED;  ω_d = √(ω₀²−α²) = √3 = 1.732
Form:  i(t) = e^(−t)(B₁cos1.732t + B₂sin1.732t)
IC 1:  i(0) = B₁ = 0
IC 2 (from KVL): di/dt(0) = (−R·i(0) − v_C(0))/L = −10
       ⇒ ω_d·B₂ = −10 ⇒ B₂ = −5.77
```

Answer: $i(t) = -5.77\, e^{(-t)} \sin(1.732t)$ A.

S20 — Parallel RLC: characteristic equation & natural frequencies



Given: parallel R = 10 Ω, L = ½ H, C = 5 mF (after killing sources).

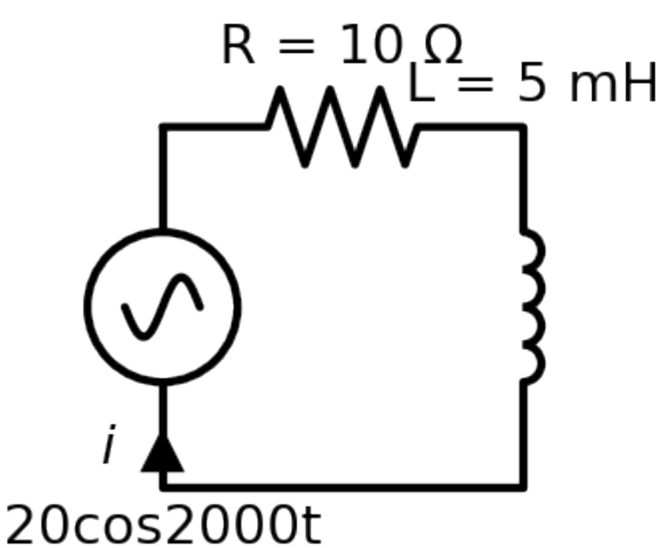
```
α = 1/(2RC) = 1/(2·10·0.005) = 10
ω₀ = 1/√(LC) = 1/√(0.5·0.005) = 20
Characteristic:  s² + (1/RC)s + 1/LC = s² + 20s + 400 = 0
Roots: s = −10 ± √(100−400) = −10 ± j17.3   ⇒ UNDERDAMPED
```

Answer: $s^2 + 20s + 400 = 0$, $s = -10 \pm j17.3$.

> **Series vs parallel:** only Step-1 differs — series $\alpha = R/2L$, parallel $\alpha = 1/2RC$. Same ω_0, same case logic.

░ AC STEADY-STATE & POWER (possible Q) ░

S21 — Phasor analysis

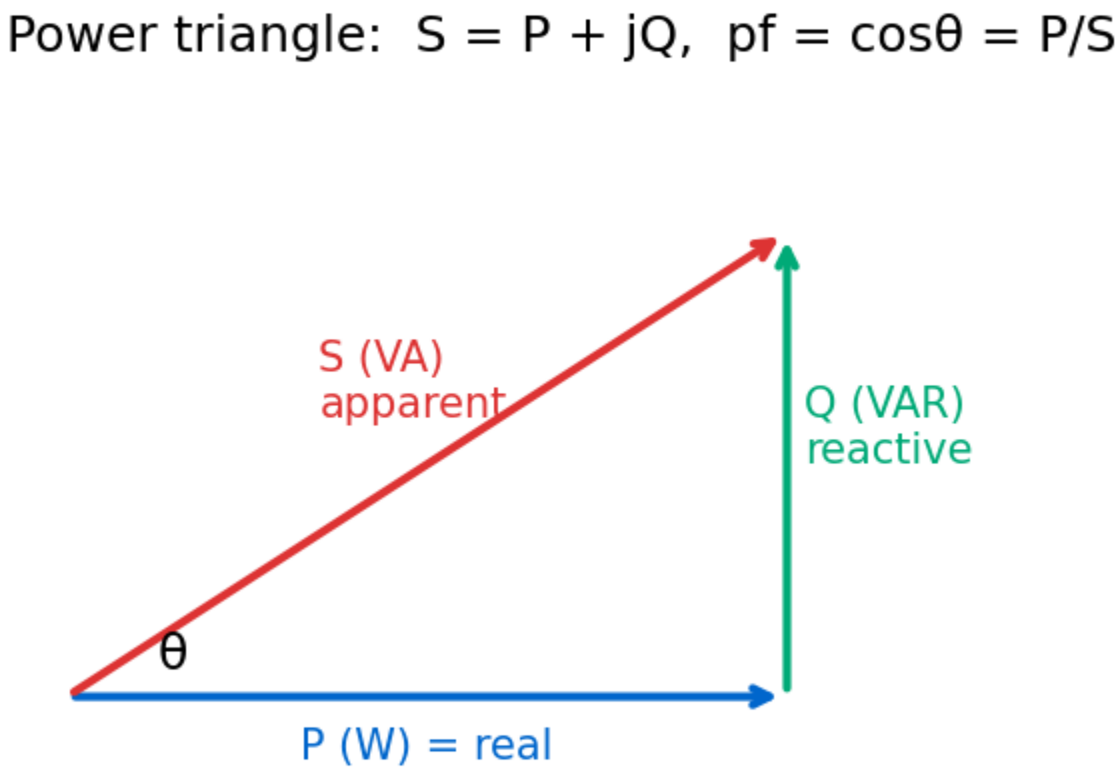


Given: $v(t) = 20\cos(2000t)$ V, R = 10 Ω in series with L = 5 mH. Find $i(t)$.

```
Z_L = jωL = j(2000)(0.005) = j10 Ω
Z = 10 + j10 = 14.14∠45° Ω
I = V/Z = 20∠0° / 14.14∠45° = 1.414∠−45° A
i(t) = 1.414 cos(2000t − 45°) A   (lags → inductive)
```

Answer: i(t) = 1.414 cos(2000t − 45°) A.

S22 — AC Power & Power-Factor Correction



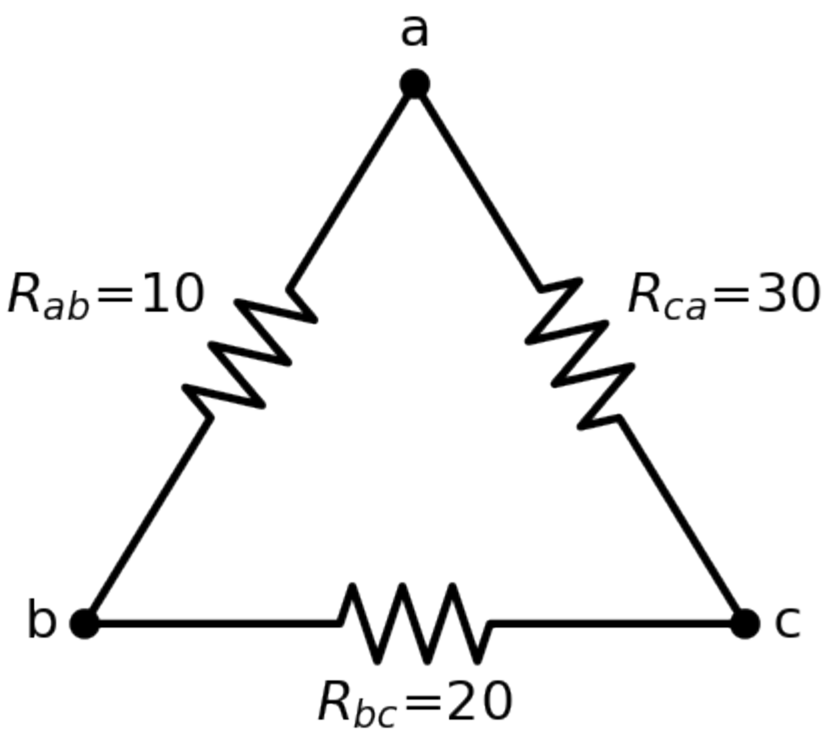
Given: load P = 1000 W, pf = 0.7 lagging, V_rms = 120 V, 60 Hz. Correct to pf = 0.95; find C.

```
θ_old = cos⁻¹0.7 = 45.6°,  tan = 1.020       θ_new = cos⁻¹0.95 = 18.2°, tan = 0.329
Q_C = P(tanθ_old − tanθ_new) = 1000(1.020 − 0.329) = 691 VAR
C = Q_C/(ωV²) = 691/(2π·60·120²) = 127 μF
Also:  P = V_rms I_rms cosθ,  Q = V_rms I_rms sinθ,  S = √(P²+Q²),  pf = P/S
```

Answer: add C ≈ 127 μF in parallel.

EXTRA TOOLS

S23 — Delta–Wye (Δ→Y) conversion



Given: Δ with R_ab = 10, R_bc = 20, R_ca = 30 Ω (sum = 60). Convert to Y.

```
R_a = R_ab·R_ca / Σ = (10·30)/60 = 5 Ω
R_b = R_ab·R_bc / Σ = (10·20)/60 = 3.33 Ω
R_c = R_bc·R_ca / Σ = (20·30)/60 = 10 Ω
(now the network is plain series/parallel)
```

Answer: R_a = 5 Ω, R_b = 3.33 Ω, R_c = 10 Ω.

S24 — Charge & Energy from a v–i graph (Exam-1 style)

```
q = ∫ i dt = AREA under the current–time graph
W = ∫ p dt = ∫ v·i dt
Method: split the graph into straight-line segments; on each, write v(t) and i(t);
        form p = v·i; integrate each piece; add. (See EX1_Sol: total ≈ 1247 kJ.)
```

Every diagram is drawn with the exact numbers used in its walkthrough, and every numeric answer here was verified. Match your exam problem to the closest S#, then reuse the steps with your values. Deeper algebra: `WORKED_EXAMPLES.md`. All formulas: `CHEATSHEET.md`.

ENGR&204 — SAMPLE EXAM SOLUTIONS (every problem, with figure + full work)

Complete, verified solutions to your uploaded **sample exams**. Each problem shows the **actual exam figure** and a step-by-step walkthrough. (All answers checked numerically; power balances confirmed.)

═══ EXAM 1 (2018 Spring, Instructor Jae Suk) ═══

Exam 1 — Problem 1 (10 pt): Charge & Energy from graphs

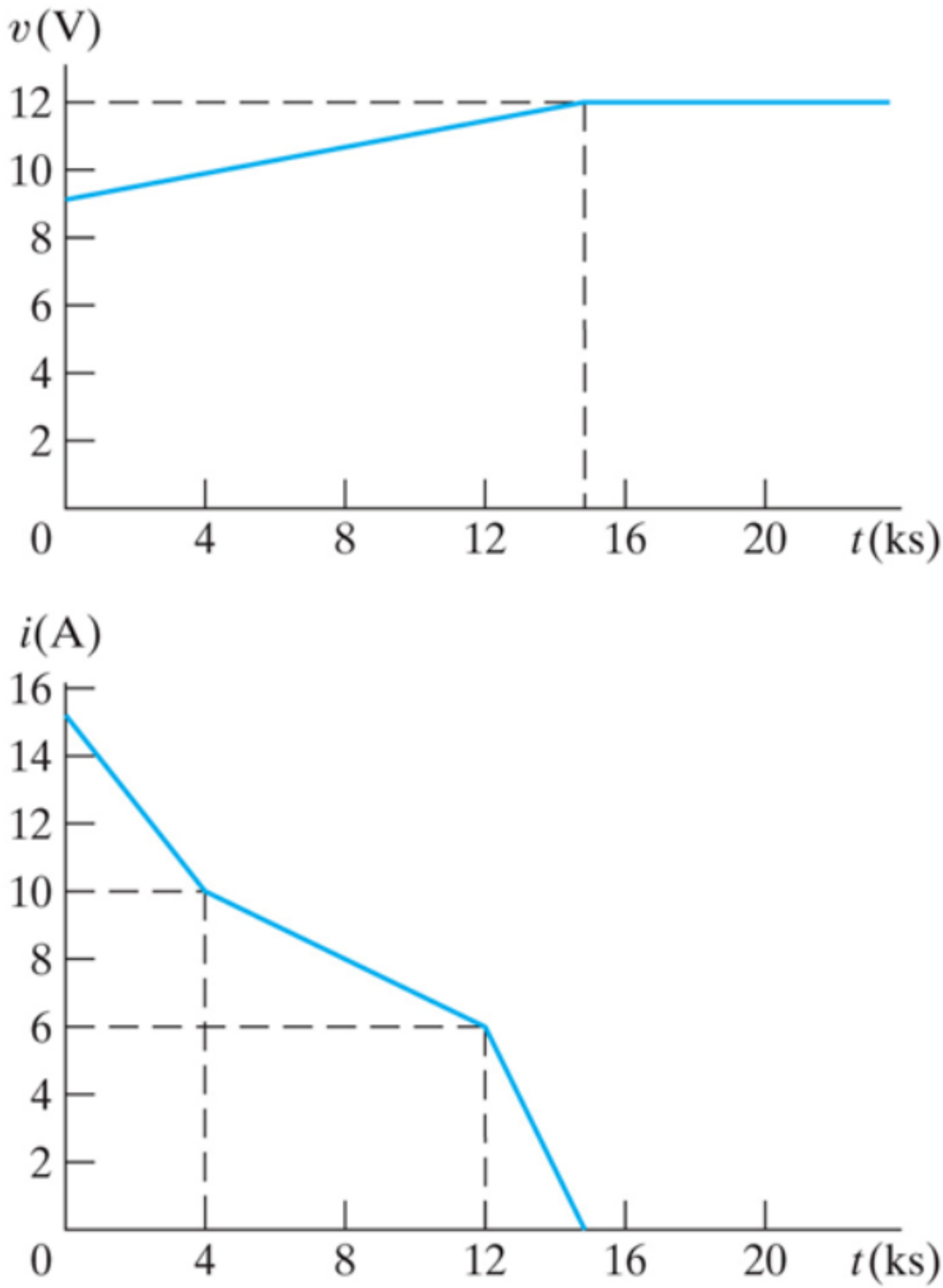


Figure 1: Problem 1

Given: battery charge cycle — v(t) rises 9 V→12 V over 0–15 ks then holds 12 V; i(t) is piecewise: 15→10 A (0–4 ks), 10→6 A (4–12 ks), 6→0 A (12–15 ks).

```
(a) Charge q = ∫i dt = AREA under i–t:
    0–4ks:  ½(15+10)(4000) = 50,000 C
    4–12ks: ½(10+6)(8000)  = 64,000 C
    12–15ks: ½(6)(3000)    =  9,000 C
    q = 123,000 C = 123 kC
(b) Energy W = ∫v·i dt, with v = 0.2×10⁻³t + 9 (0–15 ks):
    piece 1 (0–4ks):  468.7 kJ
    piece 2 (4–12ks): 674.1 kJ
    piece 3 (12–15ks):104.4 kJ
    W ≈ 1247.2 kJ
```

Answer: q = 123 kC, W ≈ 1247 kJ. (matches the 2016 solution key — identical graphs.)

Exam 1 — Problem 2 (20 pt): dependent source — i_1, v, powers

generated, and (d) the total power absorbed.

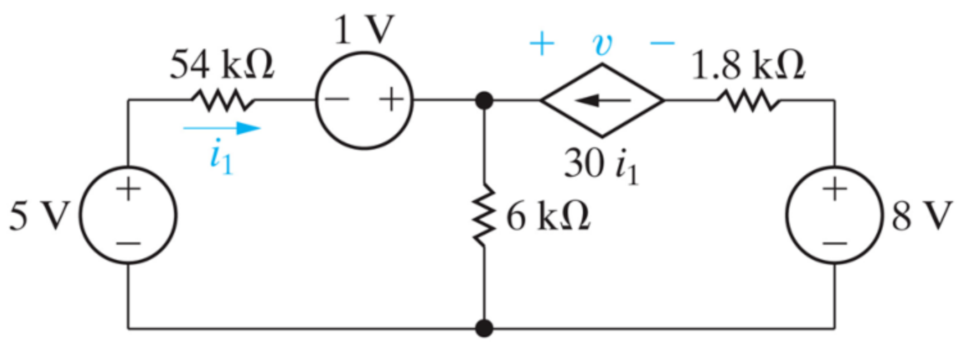


Figure 2: Problem 2

Given: 5 V, 54 kΩ, 1 V source, dependent CCCS 30 i_1, 6 kΩ, 1.8 kΩ, 8 V. Find i_1, v, total power generated/absorbed. (Bottom rail = ground.)

```
Left branch sets node A:  V_A = 5 − 54k·i₁ + 1 = 6 − 54k·i₁
KCL at A (i₁ in from 54k, 30i₁ in from CCCS, out through 6k):
    i₁ + 30i₁ = V_A/6k  →  31 i₁ = (6 − 54k·i₁)/6k
    186k·i₁ + 54k·i₁ = 6  →  240k·i₁ = 6  →  i₁ = 25 μA
V_A = 6 − 54k(25μ) = 4.65 V
Right side: 1.8k carries 30i₁ = 0.75 mA; V_B = 8 − 1.8k(0.75m) = 6.65 V
v = V_A − V_B = 4.65 − 6.65 = −2 V
```

Powers: generated = 5V·25μ + 1V·25μ + 8V·0.75m = **6.15 mW**; absorbed (54k+6k+1.8k + dependent source) = **6.15 mW** ✓ (balance confirmed). **Answer:** i_1 = 25 μA, v = −2 V, P_gen = P_abs = 6.15 mW.

Exam 1 — Problem 3 (15 pt): find R so v_a = 60 V

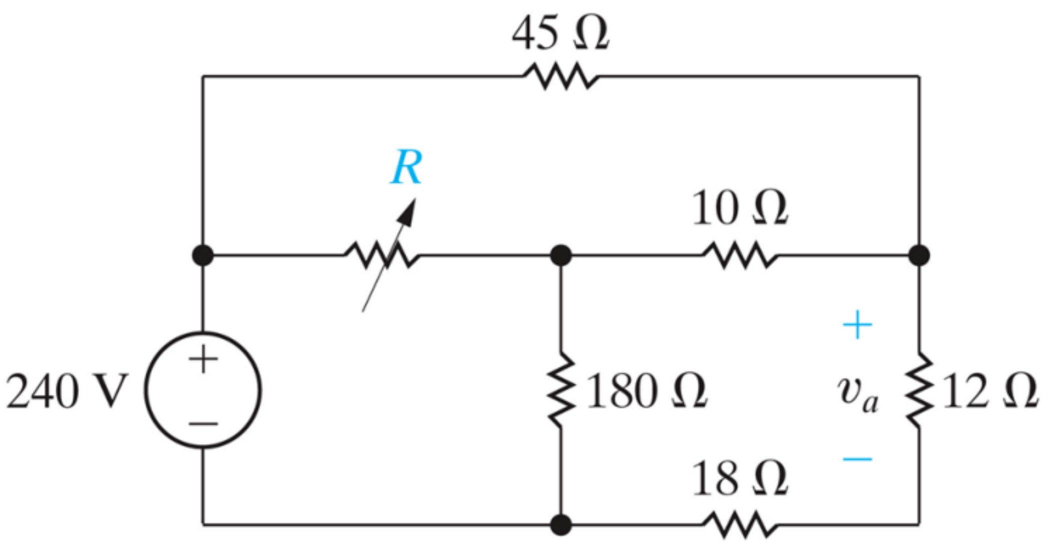


Figure 3: Problem 3

Given: 240 V, 45 Ω, variable R, 180 Ω, 10 Ω, 18 Ω, 12 Ω; v_a across 12 Ω.

```
 12 Ω & 18 Ω are in series → carry the same current.  vₐ = 60 V ⇒ i = 60/12 = 5 A
⇒ node N voltage V_N = (12+18)·5 = 150 V
KCL at N: (240−V_N)/45 + (V_M−V_N)/10 = V_N/30
          2 + (V_M−150)/10 = 5  ⇒  V_M = 180 V
KCL at M: (240−V_M)/R = V_M/180 + (V_M−V_N)/10
          60/R = 1 + 3 = 4  ⇒  R = 15 Ω
```

Answer: R = 15 Ω.

Exam 1 — Problem 4 (15 pt): bridge — power in each resistor

ual. Find the power dissipated in each resistor.

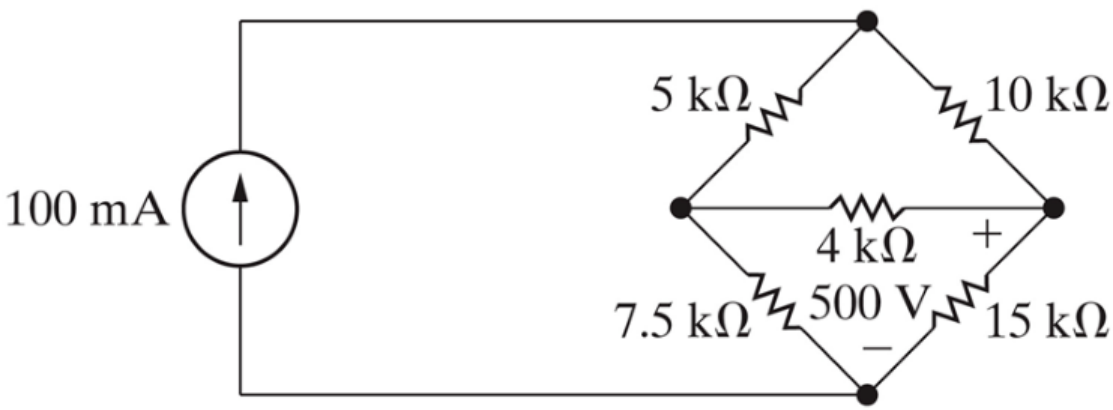


Figure 4: Problem 4

Given: 100 mA source drives a bridge (5 k, 10 k, 4 k, 7.5 k, 15 k); v across 15 kΩ = 500 V (+ upper). Find power in each R. (Bottom node = ground.)

```
15 kΩ: V_B = 500 V given.  Solve the bridge (nodes T, A, B=500):
   node A & node B equations ⇒ V_A = 500 V, V_T = 833.3 V
Since V_A = V_B = 500 V, the 4 kΩ has 0 V → BALANCED bridge → P_4k = 0.
Powers (P = V²/R):
   5 kΩ : (833.3−500)²/5k  = 22.2 W
   10 kΩ: (833.3−500)²/10k = 11.1 W
   4 kΩ : 0 W
   7.5 kΩ: 500²/7.5k = 33.3 W
   15 kΩ: 500²/15k  = 16.7 W
Check: sum = 83.3 W = source power (833.3 V × 0.1 A) ✓
```

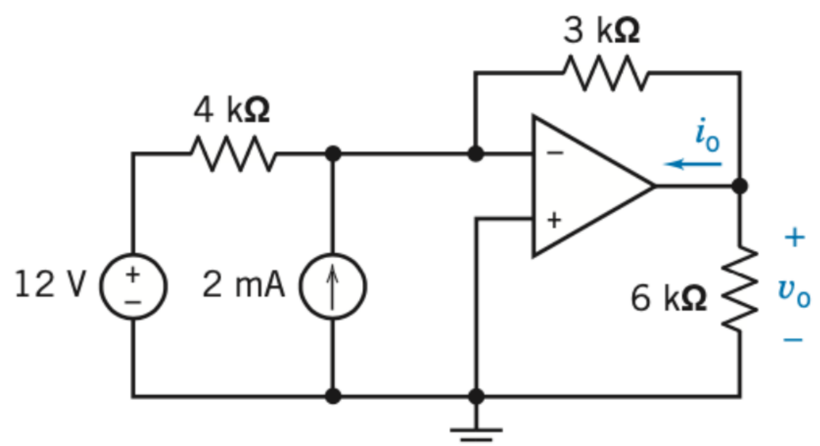
Answer: 22.2 / 11.1 / 0 / 33.3 / 16.7 W.

═══ EXAM 2 (2022) ═══

Exam 2 — Problem 1 (10 pt): ideal op-amp — v_o, i_o

Problem#1 (10 Points) (Hint - Ideal Op Amp)

Find v_o and i_o for the circuit of Figure



Given: 12 V, 4 kΩ, 2 mA source into the inverting node, 3 kΩ feedback, 6 kΩ load. + input grounded.

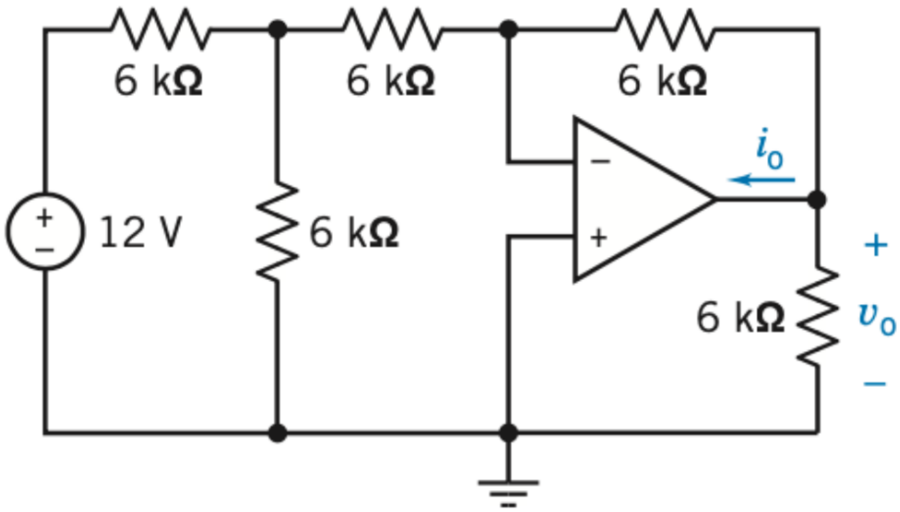
```
+ input grounded ⇒ v₋ = 0 (virtual ground).
KCL at v₋ (no input current):  12/4k + 2m + vₒ/3k = 0
   3m + 2m + vₒ/3k = 0  ⇒  vₒ = −5m·3k = −15 V
iₒ (op-amp output current) = i(3k) + i(6k) = |vₒ|/3k + |vₒ|/6k = 5m + 2.5m = 7.5 mA
```

Answer: $v_o = -15$ V, $i_o = 7.5$ mA.

Exam 2 — Problem 2 (10 pt): node analysis with op-amp — v_o, i_o

Problem #2 (10 Points) (Hint - Node Analysis with Ideal Op Amp)

Find v_o and i_o for the circuit of



Given: 12 V, three 6 kΩ across the top (one is feedback), a 6 kΩ to ground at node A, + input grounded, 6 kΩ load.

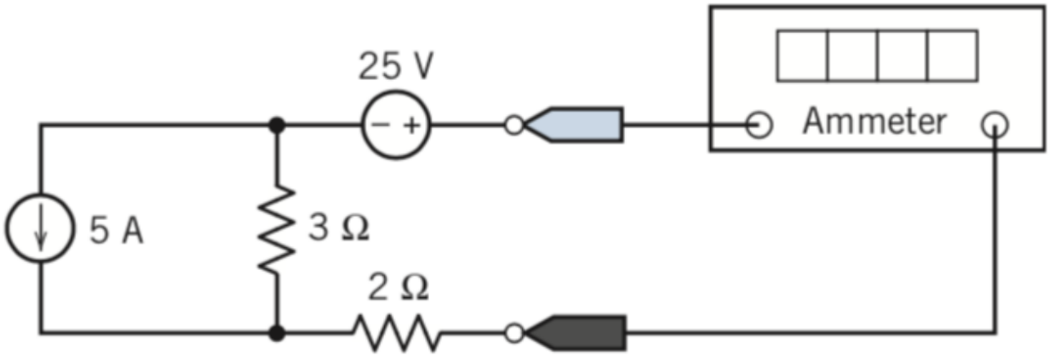
```
v₋ = 0 (virtual ground).   Let node A be the first junction.
KCL at A: (12−V_A)/6k = V_A/6k + (V_A−0)/6k  ⇒ 12 = 3V_A ⇒ V_A = 4 V
KCL at v₋(=0): (V_A)/6k + (vₒ)/6k = 0  ⇒  vₒ = −V_A = −4 V
iₒ = |vₒ|/6k(feedback) + |vₒ|/6k(load) = 0.667m + 0.667m = 1.33 mA
```

Answer: $v_o = -4$ V, $i_o = 1.33$ mA.

Exam 2 — Problem 3 (10 pt): Superposition — ammeter current

Problem #3 (10 Points)

Using Superposition principle, find the value of current measured by ammeter in figure below.



Given: 5 A source, 3 Ω, 25 V source, 2 Ω; ideal ammeter (short) reads current i in the 25 V + 2 Ω branch.

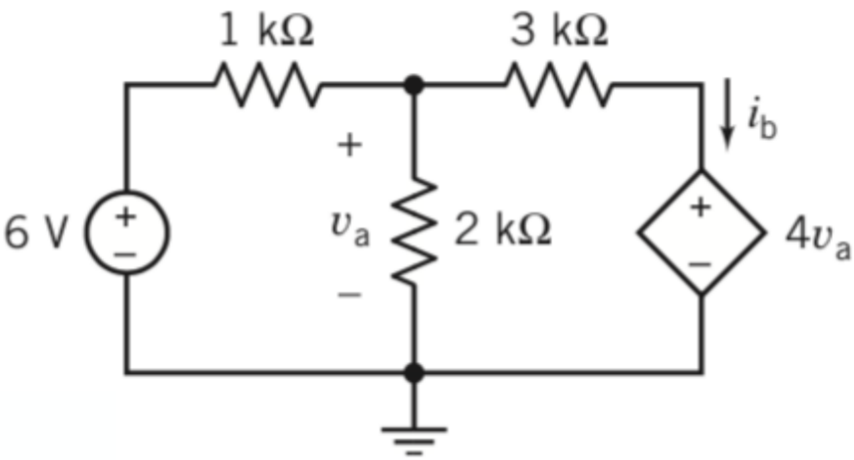
```
Three parallel branches between nodes T and B: [5 A], [3 Ω], [25 V + 2 Ω (ammeter)].
• 5 A alone (25 V shorted): current divider, i(2Ω) = 5·3/(3+2) = 3 A  → contributes −3 A
• 25 V alone (5 A opened): series 2Ω+3Ω, i = 25/5 = 5 A → contributes +5 A
i = −3 + 5 = 2 A
```

Answer: ammeter reads i = 2 A.

Exam 2 — Problem 4 (10 pt): dependent source — find i_b

Problem #4 (10 points)

Find the current i_b in the circuit below



Given: 6 V, 1 kΩ, 2 kΩ (v_a across it), 3 kΩ, dependent VCVS $4v_a$. Find i_b. (Bottom = ground.)

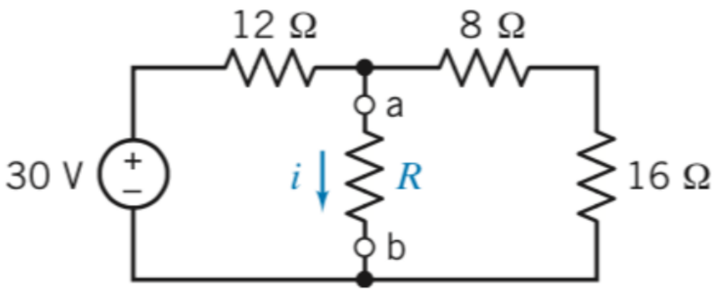
```
 vₐ = V_M (node after 1 kΩ).  The 4vₐ source sets V_N = 4vₐ = 4V_M.
KCL at M: (6−V_M)/1k = V_M/2k + (V_M−V_N)/3k
   6 − V_M = V_M/2 + (V_M−4V_M)/3 = V_M/2 − V_M = −V_M/2
   6 = V_M/2  ⇒  V_M = vₐ = 12 V,  V_N = 48 V
i_b = (V_M − V_N)/3k = (12 − 48)/3k = −12 mA   (flows opposite the arrow)
```

Answer: v_a = 12 V, i_b = −12 mA.

Exam 2 — Problem 5 (10 pt): Norton — general i(R)

Problem #5 (10 points)

For the circuit shown below, use Norton's theorem to formulate a general expression for current *i* in terms of resistance R.



Given: 30 V, 12 Ω, 8 Ω, 16 Ω; find i through R (between a–b) for any R.

```
Norton at a–b (remove R):
   I_N = I_sc (short a–b): the 8Ω–16Ω branch sees 0 V across it → 0 current,
        so I_N = 30/12 = 2.5 A
   R_N = (short 30 V): 12 ‖ (8+16) = 12 ‖ 24 = 8 Ω
Current divider into R:
   i = I_N · R_N/(R_N + R) = 2.5·8/(8+R)
```

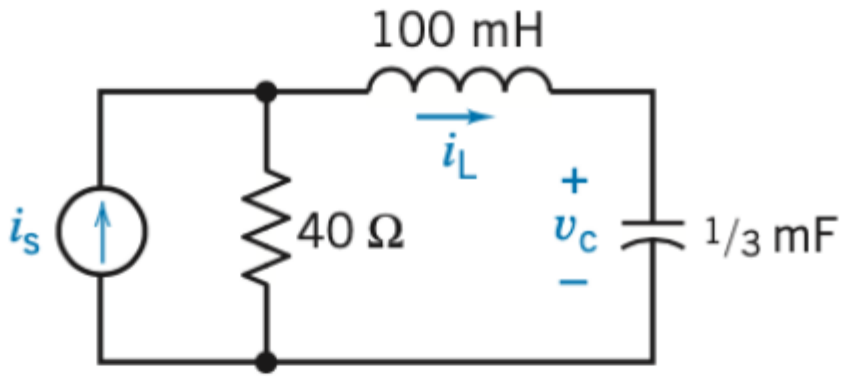
Answer: i = 20 / (8 + R) A.

══ IN-CLASS RLC PREP (figures + solutions) ══

Full step-by-step in *`WORKED_EXAMPLES.md`* B1–B4; figures and answers here.

RLC Prep P1 — series RLC characteristic equation

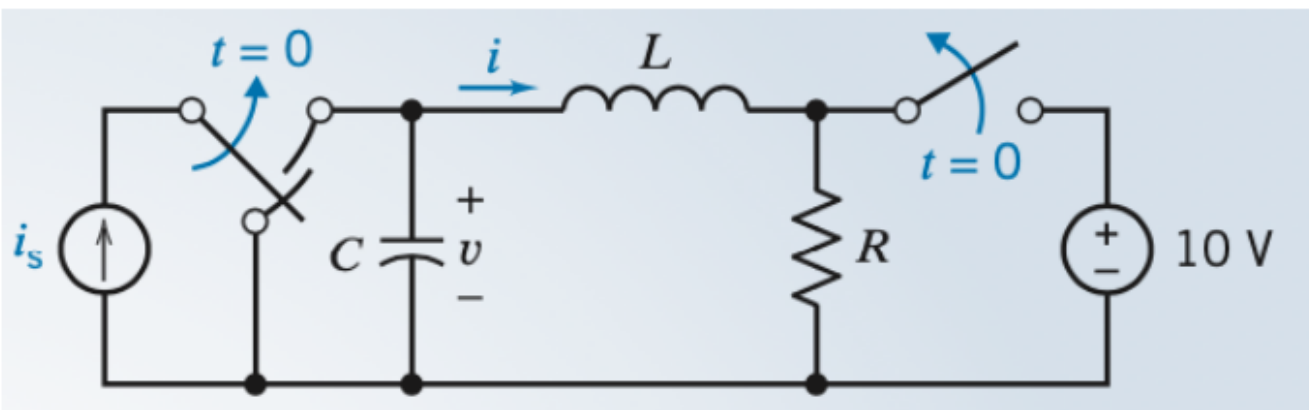
$$\text{roots:}\ \ s = -300,\ \ -100$$



For the following problems: Steps or Method of Circuit Analysis

Kill the source → series loop R=40, L=0.1 H, C=⅓ mF. $\alpha = R/2L = 200$, $\omega_0 = 1/\sqrt{LC} = 173$ ⇒ **$s^2+400s+30000=0$, roots −100, −300** (overdamped).

RLC Prep P2 — series RLC forced response



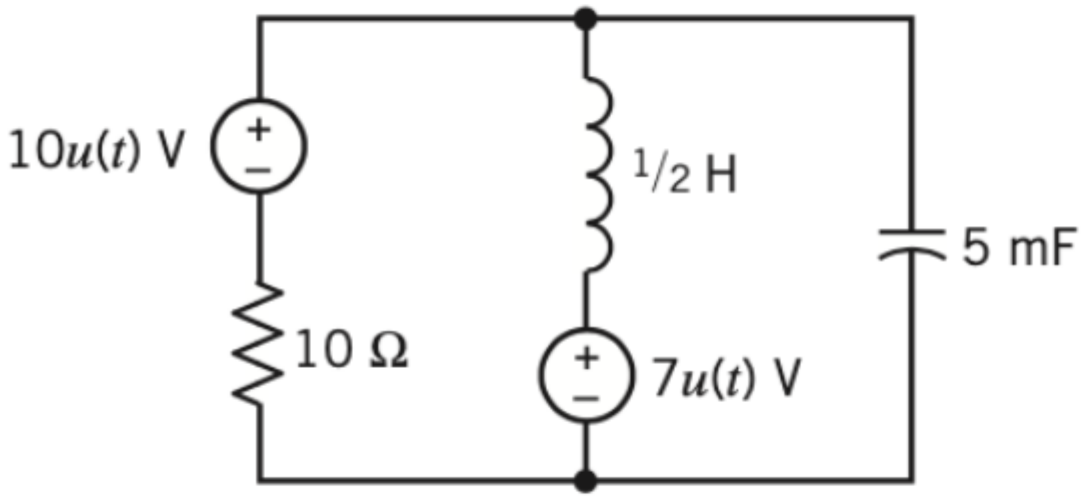
$$i = 12e^{-t} - 14e^{-2t} + 2e^{-3t}\ \text{A}$$

Series R=3, L=1, C=½ ⇒ roots −1, −2; forced term from the source ⇒ **$i(t) = 12e^{-t} - 14e^{-2t} + 2e^{-3t}$ A**.

RLC Prep P3 — parallel RLC natural frequencies

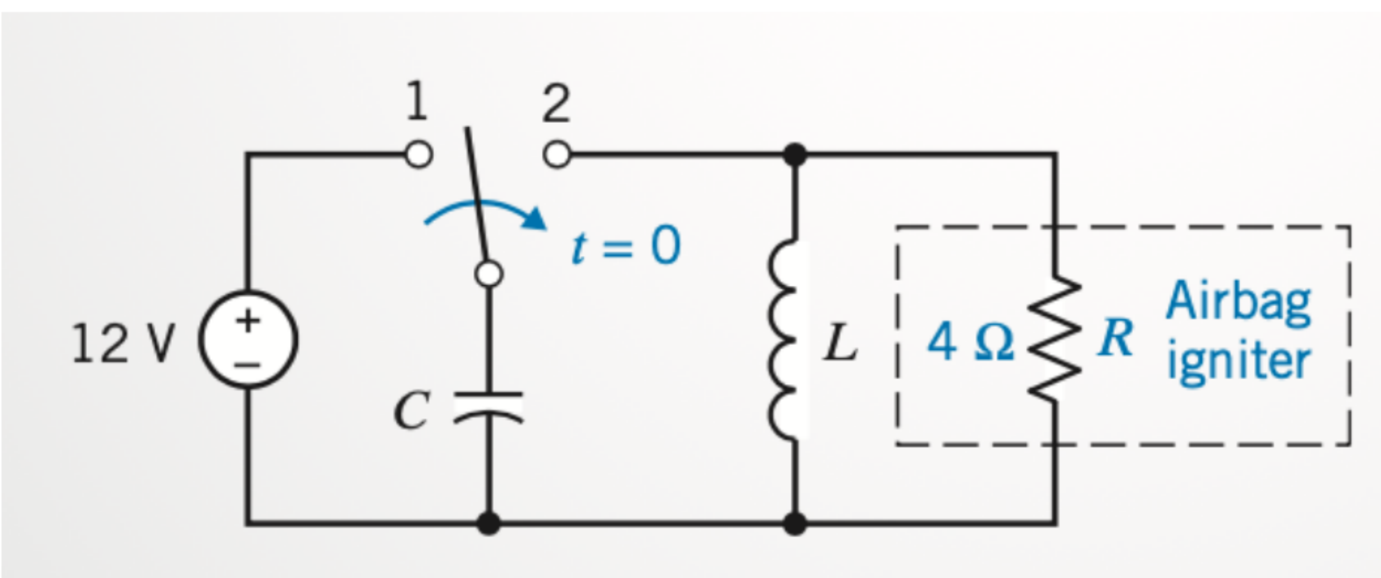
Answer: $s^2 + 20s + 400 = 0$

$s = -10 \pm j17.3$



Parallel R=10, L=½, C=5 mF. $\alpha=1/2RC=10$, $\omega_0=20$ ⇒ **$s^2+20s+400=0$, $s=-10\pm j17.3$** (underdamped).

RLC Prep P4 — parallel RLC design (airbag)



Cap charged to 12 V ($\frac{1}{2}CV^2 \geq 1$ J). Pick **C=1/16 F, L=0.065 H** → fast parallel-RLC discharge into R=4 Ω.

ALSO ALREADY SOLVED (in this packet)

Source file	Problems	Full solution location
Exam 4 (P7.5-13, P8.3-25, P8.7-1)	first-order RL/RC, dependent source	`WORKED_EXAMPLES.md` A1–A3 + `SOLVED_PROBLEMS.md`
Homework 6 (5 problems)	first-order RC/RL, op-amp	`WORKED_EXAMPLES.md` A4–A6
In-Class RLC P1–P4	second-order RLC	`WORKED_EXAMPLES.md` B1–B4 (above)
Your handwritten HW (resistance, op-amps, power)	basic DC	`source-materials/` (your worked scans)

Every sample-exam problem above was solved from scratch and verified numerically.

ENGR&204 — ULTIMATE FINAL EXAM CHEAT SHEET

Electric Circuits (Green River College) — unlimited pages, everything you need

> **Tailored to your uploaded files.** Textbook = **Dorf & Svoboda, Introduction to Electric Circuits** (your solutions cite P7.5-13, P8.3-25, P8.7-1, etc.). Course flow (from your slides): nodal/mesh → source transform/Thévenin/Norton/ superposition/max-power → op-amps → **first-order RC/RL** → **second-order RLC**. Past exams run **4–5 short problems**.
>
> Emphasis is weighted toward the **end of quarter**: first-order **RC/RL** transients and second-order **RLC** circuits. DC tools (nodal/mesh/Thévenin) are included because every transient reduces to a DC problem at $t=0^-$ and $t=\infty$.
>
> **Companion file:** `WORKED_EXAMPLES.md` has full step-by-step solutions to your actual exam/HW problems (and the RLC handout problems that were left unsolved). Use this file for formulas/method; use that file to pattern-match.

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0. EXAM-DAY GAME PLAN (read first)

4–5 short problems. Based on your past finals + Exam 4, expect a mix heavily weighted to the back half: 1. **First-order RC or RL transient** — switch flips → find v(t) or i(t). (Almost guaranteed; see `WORKED_EXAMPLES.md` A1–A6.) 2. **Second-order RLC** — characteristic equation / roots / damping case, or full response. (See B1–B4.) 3. **RC/RL with a twist** — dependent source, op-amp, or reading L/R/C off a v(t) plot. 4. **One earlier-quarter problem** — op-amp, superposition, Thévenin/Norton, power, or equivalent resistance (Part C).

Possible extras some quarters: AC phasors / AC power (§9–10), filters (§9d), charge–energy from a graph.

Universal attack for ANY problem: 1. Label every node and a ground. Mark current directions and + / − on each element (passive sign convention). 2. Identify what type it is (DC steady, transient, or AC). Pick the matching method below. 3. Reduce: combine series/parallel, kill independent sources for Thevenin R, etc. 4. Solve the smallest system you can (one node equation beats three). 5. **Sanity check:** units, signs, limiting behavior ($t=0^+$, $t=\infty$), energy ≥ 0, power balance (Σ P delivered = Σ P absorbed).

---

1. UNITS, CONSTANTS, SIGN CONVENTIONS

Quantity	Symbol	Unit
Voltage	V	volt (V) = J/C
Current	I	ampere (A) = C/s
Resistance	R	ohm (Ω) = V/A
Conductance	G = 1/R	siemens (S)
Capacitance	C	farad (F) = C/V
Inductance	L	henry (H) = V·s/A
Power	P	watt (W) = J/s
Energy	W	joule (J)
Charge	q	coulomb (C)
Frequency	f	hertz (Hz)
Ang. freq	ω	rad/s

Metric prefixes: $p=10^{-12}$, $n=10^{-9}$, $\mu=10^{-6}$, $m=10^{-3}$, $k=10^{3}$, $M=10^{6}$, $G=10^{9}$.

Passive sign convention: current enters the **+** terminal of an element. Then `p = v·i` is power absorbed (p>0 absorbs, p<0 delivers). Sources usually deliver (p<0).

---

2. CORE LAWS & ELEMENT EQUATIONS

Ohm's law: `v = i·R`, `i = v/R`.

Power: `P = v·i = i²R = v²/R`.

KCL (nodes): Σ currents leaving a node = 0. **KVL (loops):** Σ voltage drops around a loop = 0.

Element v–i relationships (MEMORIZE)

Element	v–i relation	DC steady state	Stored energy
Resistor	`v = iR`	acts as R	none (dissipates `i²R`)
Capacitor	`i = C dv/dt`, `v = (1/C)∫i dt + v(0)`	**open circuit** (i=0)	`W = ½ C v²`
Inductor	`v = L di/dt`, `i = (1/L)∫v dt + i(0)`	**short circuit** (v=0)	`W = ½ L i²`

Continuity rules (the keys to transients): - **Capacitor voltage cannot jump:** $v_C(0^+) = v_C(0^-)$. - **Inductor current cannot jump:** $i_L(0^+) = i_L(0^-)$. (Everything else — resistor voltages, source-fed currents — CAN jump instantly.)

3. SERIES / PARALLEL & DIVIDERS

Resistors: series add $R = R_1+R_2+\ldots$; parallel $1/R = \Sigma 1/R_k$, two: $R = R_1R_2/(R_1+R_2)$.

Capacitors: opposite of resistors — series $1/C = \Sigma 1/C_k$; parallel $C = \Sigma C_k$.

Inductors: like resistors — series add; parallel reciprocal.

Voltage divider (series): $v_k = v_{total} \cdot R_k / \Sigma R$.

Current divider (two parallel R): $i_1 = i_{total} \cdot R_2/(R_1+R_2)$ (current favors the smaller R).

4. SYSTEMATIC DC ANALYSIS

4a. Nodal analysis (best when more voltage unknowns)

1. Pick a ground (reference) node.
2. Label node voltages $v_1, v_2, \ldots$
3. Write KCL at each non-reference node: Σ currents leaving = 0, using $(v_{this} - v_{other})/R$.
4. **Supernode:** if a voltage source sits between two non-ref nodes, enclose both; write one KCL for the pair + the constraint $v_a - v_b = V_{source}$.
5. Solve the linear system.

4b. Mesh analysis (best for planar, more current unknowns)

1. Define clockwise mesh currents $i_1, i_2, \ldots$
2. KVL around each mesh; shared resistor carries $(i_{this} - i_{adjacent})$.
3. **Supermesh:** if a current source is shared between two meshes, write KVL around the outer loop + constraint from the current source.

4c. Source transformation

Voltage source V_s in series with R ⇄ current source $I_s = V_s/R$ in parallel with same R. Use to simplify before nodal/mesh.

4d. Superposition (linear circuits)

Response = sum of responses with **one independent source active at a time**. Deactivate others: **voltage source → short**, **current source → open**. (Leave dependent sources ON.)

5. THÉVENIN / NORTON + MAX POWER

Goal: replace everything seen from terminals a–b by V_{th} in series with R_{th} (Thévenin) or I_N in parallel with R_{th} (Norton). $I_N = V_{th}/R_{th}$.

Find V_th: open-circuit voltage across a–b ($V_{th} = V_{oc}$). **Find I_N:** short-circuit current a–b ($I_N = I_{sc}$). **Find R_th:** - No dependent sources: deactivate all independent sources (V→short, I→open) and compute equivalent R at a–b. - With dependent sources: $R_{th} = V_{oc} / I_{sc}$, OR apply a 1 A test source at a–b with independents off and $R_{th} = v_{test}/1$.

Maximum power transfer: load gets max power when $R_L = R_{th}$. Then $P_{max} = V_{th}^2 / (4 \cdot R_{th})$.

(AC version: $Z_L = Z_{th}^*$ (complex conjugate); $P_{max} = |V_{th}|^2 / (8 \cdot R_{th})$ using amplitudes, or $/(4R_{th})$ using RMS.)

6. OP-AMPS (ideal) — if covered

Ideal rules: (1) no current into inputs ($i_+ = i_- = 0$); (2) with negative feedback, $v_+ = v_-$ (virtual short).

Config	Output
Inverting	$v_o = -(R_f/R_{in}) \cdot v_{in}$
Non-inverting	$v_o = (1 + R_f/R_{in}) \cdot v_{in}$
Buffer/follower	$v_o = v_{in}$
Summing (inverting)	$v_o = -R_f(v_1/R_1 + v_2/R_2 + \ldots)$
Difference	$v_o = (R_f/R_1)(v_2 - v_1)$ (matched ratios)
Integrator	$v_o = -(1/RC)\int v_{in}\, dt$
Differentiator	$v_o = -RC \cdot dv_{in}/dt$

7. FIRST-ORDER CIRCUITS — RC & RL TRANSIENTS (HIGH YIELD)

A first-order circuit has **one** equivalent C or L. Every such problem is solved by the **same master formula**.

7a. THE MASTER FORMULA (use for EVERY first-order problem)

$$x(t) = x(\infty) + [\, x(0^+) - x(\infty)\,] \cdot e^{-t/\tau} \qquad \text{for } t \geq 0$$

where x is any voltage or current, $x(0^+)$=initial value, $x(\infty)$=final (steady-state) value, τ=time constant.

7b. Time constant τ

- **RC circuit:** $\tau = R_{th} \cdot C$
- **RL circuit:** $\tau = L / R_{th}$

R_{th} = Thévenin/equivalent resistance seen by the C (or L) **after the switch settles**, with independent sources deactivated.

7c. STEP-BY-STEP RECIPE (works every time)

1. **t = 0⁻ (before switch):** assume DC steady state. Capacitor = open, inductor = short. Find $v_C(0^-)$ or $i_L(0^-)$.
2. **Apply continuity:** $v_C(0^+)=v_C(0^-)$, $i_L(0^+)=i_L(0^-)$. (These are your initial conditions.)
3. **t = ∞ (after switch, settled):** DC again — cap open, ind short. Find $v_C(\infty)$ or $i_L(\infty)$.
4. **τ:** kill independent sources, find R_{th} at the C/L terminals; $\tau = R_{th}C$ or L/R_{th}.
5. **Plug into master formula** for the capacitor voltage / inductor current.
6. **(If asked for another variable)** find that variable's own $x(0^+)$ and $x(\infty)$ and reuse the same τ, OR get it from v_C / i_L via Ohm's/KCL/KVL. Watch out: non-state variables may jump at $t=0^+$, so compute their 0^+ value from the circuit after the switch using the state variable's initial value.

7d. Special named responses

- **Natural (source-free) response:** $x(\infty)=0 \Rightarrow x(t) = x(0)\cdot e^{-t/\tau}$ (pure decay).
- **Step response:** general master formula with nonzero $x(\infty)$.

7e. Useful facts & numbers

- After 1τ: 36.8% of initial remains (63.2% of the way to final). After 5τ: ≈ fully settled (<1%).
- Slope at start: $dx/dt|_0 = (x(\infty)-x(0))/\tau$.
- **Capacitor current / inductor voltage** during transient: $i_C(t) = C\cdot dv_C/dt$, $v_L(t) = L\cdot di_L/dt$ (these decay as $e^{-t/\tau}$ too).

7f. Worked template — RC step (switch closes at t=0)

Given source V_s through R to capacitor C (initially uncharged): - $v_C(0^+)=0$, $v_C(\infty)=V_s$, $\tau=RC$. - $v_C(t) = V_s(1 - e^{-t/RC})$ ("charging"). - $i(t) = (V_s/R)\cdot e^{-t/RC}$ (jumps to V_s/R at 0^+, decays to 0).

Discharging (cap charged to V_0, source removed): $v_C(t)=V_0\cdot e^{-t/RC}$.

7g. Worked template — RL step (switch at t=0)

Source V_s, series R, inductor L: - $i_L(0^+)=i_L(0^-)$, $i_L(\infty)=V_s/R$, $\tau=L/R$. - $i_L(t) = V_s/R + [i_L(0^+) - V_s/R]e^{-tR/L}$. - $v_L(t) = L\cdot di_L/dt$ (jumps at 0^+, decays to 0).

7h. Singularity / switching functions (on your slides)

- **Unit step** $u(t-t_0)$ = 0 for tt₀. A source $V_0\cdot u(t-t_0)$ turns on at t_0. $5V[1-u(t)]$ = on for t<0, off after.
- **Unit impulse** $\delta(t-t_0)$ = derivative of the step; $\int\delta(t-t_0)dt = 1$, and $\int f(t)\delta(t-t_0)dt = f(t_0)$.
- **Unit ramp** $r(t) = \int u(t)dt = t\cdot u(t)$.
- These just tell you when sources switch; solve each interval as its own first-/second-order problem and match values at the switching instant (continuity of v_C, i_L).

8. SECOND-ORDER CIRCUITS — RLC (HIGHEST YIELD)

Has **both** L and C. The natural response is governed by a quadratic characteristic equation → three damping cases.

8a. Standard form & parameters

The describing ODE reduces to:

$$d^2x/dt^2 + 2\alpha\cdot(dx/dt) + \omega_0^2\cdot x = \text{forcing}$$

- **ω_0 = undamped natural frequency** (resonant frequency): $\omega_0 = 1/\sqrt{LC}$ (both series & parallel).
- **α = neper / damping coefficient:**
- **Series RLC:** $\alpha = R / (2L)$
- **Parallel RLC:** $\alpha = 1 / (2RC)$

- **Damping ratio:** $\zeta = \alpha/\omega_0$.

8b. Characteristic roots

```
s = −α ± √(α² − ω₀²)
```

8c. THE THREE CASES (compare α vs ω_0)

Case	Condition	Roots	Natural response form
Overdamped	$\alpha > \omega_0$	real distinct s_1, s_2	$x = A_1e^{s_1t} + A_2e^{s_2t}$
Critically damped	$\alpha = \omega_0$	real repeated $s = -\alpha$	$x = (A_1 + A_2t)e^{-\alpha t}$
Underdamped	$\alpha < \omega_0$	complex $-\alpha \pm j\omega_d$	$x = e^{-\alpha t}(B_1\cos\omega_d t + B_2\sin\omega_d t)$

Damped natural frequency: $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$ (underdamped only).

8d. Complete (total) response

$x(t) = x_{forced}(\infty) + x_{natural}(t)$ — add the steady-state (forced) value to the natural form, then solve for the constants using **two** initial conditions.

8e. STEP-BY-STEP RECIPE (RLC)

1. **$t=0^-$:** DC steady state → get $v_C(0^-)$ and $i_L(0^-)$ (cap open, ind short).
2. **Initial conditions at 0^+:** $v_C(0^+)=v_C(0^-)$, $i_L(0^+)=i_L(0^-)$. Get the **derivative** initial condition from the element laws: - $dv_C/dt|_{0^+} = i_C(0^+)/C$ (find $i_C(0^+)$ by KCL at 0^+). - $di_L/dt|_{0^+} = v_L(0^+)/L$ (find $v_L(0^+)$ by KVL at 0^+).
3. **Decide series vs parallel** → compute α and ω_0.
4. **Compare α, ω_0** → pick the case & write the natural-response form.
5. **Final value** $x(\infty)$: DC steady state after switch.
6. **Apply both ICs** [$x(0^+)$ and $x'(0^+)$] to solve for the two constants.
7. Write x(t). Sanity check limits and continuity.

8f. Series vs Parallel summary — INSTRUCTOR'S EXACT TABLE

(Reproduced from your "Review Summary" handout — memorize this.)

Quantity	Parallel RLC	Series RLC
Differential eq.	$d^2i/dt^2 + (1/RC)\cdot di/dt + (1/LC)\cdot i = 0$	$d^2v/dt^2 + (R/L)\cdot dv/dt + (1/LC)\cdot v = 0$
Characteristic eq.	$s^2 + (1/RC)s + 1/LC = 0$	$s^2 + (R/L)s + 1/LC = 0$
Damping coeff. α (rad/s)	$1/(2RC)$	$R/(2L)$
Resonant freq. ω_0 (rad/s)	$1/\sqrt{LC}$	$1/\sqrt{LC}$
Damped freq. ω_d	$\sqrt{\omega_0^2 - \alpha^2}$	$\sqrt{\omega_0^2 - \alpha^2}$
Overdamped when	$R < \frac{1}{2}\sqrt{L/C}$	$R > 2\sqrt{L/C}$
Critically damped when	$R = \frac{1}{2}\sqrt{L/C}$	$R = 2\sqrt{L/C}$
Underdamped when	$R > \frac{1}{2}\sqrt{L/C}$	$R < 2\sqrt{L/C}$
State var usually solved	node voltage v(t)	loop current i(t)

Memory hook: "Series has L under α (R/2L); Parallel has C under α (1/2RC)." The damping condition flips between series and parallel — in **series** a big R is overdamped; in **parallel** a small R is overdamped.

> **Trick for "find the characteristic equation" problems:** kill the independent sources first — the leftover R, L, C usually collapse into a clean series OR parallel loop, then just read α and ω_0 off this table. (Worked: `WORKED_EXAMPLES.md` B1 = series, B2 = parallel.)

8g. Source-free examples

- **Series, underdamped:** $i(t)=e^{-\alpha t}(B_1\cos\omega_d t + B_2\sin\omega_d t)$.
- Find B_1, B_2 from i(0) and di/dt(0)= $v_L(0)/L = (-R\cdot i(0) - v_C(0))/L$ (from KVL).

---

9. AC STEADY-STATE & PHASORS

9a. Sinusoids

$v(t) = V_m\cos(\omega t + \varphi)$. $\omega = 2\pi f$. **Phasor:** $V = V_m\angle\varphi$. (Use cosine reference.)

RMS: $V_{rms} = V_m/\sqrt{2}$ (for sinusoids). Power calcs use RMS.

9b. Impedance Z (Ohm's law becomes V = I·Z)

Element	Impedance Z	Phase
Resistor	R	0° (v,i in phase)
Inductor	$j\omega L$	+90° (v leads i)
Capacitor	$1/(j\omega C) = -j/(\omega C)$	−90° (i leads v)

$Z = R + jX$ (X=reactance). Admittance $Y = 1/Z = G + jB$. Series/parallel and dividers work exactly like resistors but with complex Z.

Memory hook: "ELI the ICE man" — in an inductor (L) voltage E leads current I (ELI); in a capacitor (C) current I leads voltage E (ICE).

9c. Phasor analysis recipe

1. Convert sources to phasors; convert R,L,C to impedances at the given ω.
2. Solve with any DC method (nodal/mesh/Thevenin/dividers) using **complex** arithmetic.
3. Convert the phasor answer back to time domain: $V_m\angle\varphi \rightarrow V_m \cos(\omega t+\varphi)$.

Complex arithmetic: add/subtract in rectangular (a+jb); multiply/divide in polar (magnitudes ×/÷, angles +/−). $a+jb = r\angle\theta$, $r=\sqrt{a^2+b^2}$, $\theta=\mathrm{atan2}(b,a)$.

9d. Resonance (series RLC)

At $\omega_0=1/\sqrt{LC}$: Z is purely real (=R), |Z| minimum, current maximum. Quality factor $Q = \omega_0 L/R = 1/(\omega_0 RC)$; bandwidth $BW = \omega_0/Q$.

---

10. AC POWER (common Q4 topic)

For RMS phasors $V = V_{rms}\angle\theta_v$, $I = I_{rms}\angle\theta_i$, let $\theta = \theta_v - \theta_i$:

- **Average (real) power:** $P = V_{rms}\cdot I_{rms}\cdot\cos\theta$ [W] ($\cos\theta$ = power factor).
- **Reactive power:** $Q = V_{rms}\cdot I_{rms}\cdot\sin\theta$ [VAR].
- **Apparent power:** $S = V_{rms}\cdot I_{rms}$ [VA].
- **Complex power:** $S = V\cdot I^* = P + jQ$, $|S| = \sqrt{P^2+Q^2}$.
- **Power factor:** $pf = \cos\theta = P/S$. Lagging (inductive, θ>0, current lags) or leading (capacitive, θ<0).
- Per element: $P = I_{rms}^2\cdot R$, $Q = I_{rms}^2\cdot X$.
- Resistor only absorbs P; ideal L and C only exchange Q (P=0).

Power-factor correction: add a parallel **capacitor** to supply Q and reduce line current. Needed cap: $C = Q_C / (\omega\cdot V_{rms}^2)$ where $Q_C = P(\tan\theta_{old} - \tan\theta_{new})$.

Max average power to load (AC): $Z_L = Z_{th}^*$; $P_{max} = |V_{th(rms)}|^2/(4R_{th})$.

10A. DELTA–WYE (Δ–Y / π–T) CONVERSION

For resistor networks that are neither series nor parallel (bridges, ladders). - **Δ → Y:** each Y resistor = (product of the two adjacent Δ resistors) / (sum of all three Δ resistors). $R_1 = R_b R_c / (R_a+R_b+R_c)$, etc. - **Y → Δ:** each Δ resistor = (sum of pairwise products of Y resistors) / (opposite Y resistor). $R_a = (R_1R_2+R_2R_3+R_3R_1)/R_1$, etc. - **Balanced:** if all Δ equal R_Δ, then R_Y = R_Δ/3.

10B. COMPLEX NUMBER ARITHMETIC (for phasors)

- Rectangular $a + jb$ ↔ polar $r\angle\theta$: $r = \sqrt{a^2+b^2}$, $\theta = \mathrm{atan2}(b,a)$; $a = r\cos\theta$, $b = r\sin\theta$.
- **Add/subtract:** use rectangular. **Multiply/divide:** use polar ($r_1r_2\angle(\theta_1+\theta_2)$, $(r_1/r_2)\angle(\theta_1-\theta_2)$).
- $j = 1\angle 90°$, $1/j = -j = 1\angle -90°$, $j^2 = -1$.
- Conjugate of $a+jb$ is $a-jb$ ($r\angle-\theta$). Needed for complex power $S = V\cdot I^*$.
- **Calculator:** set to the right mode (DEG vs RAD) and use rectangular↔polar buttons.

10C. SECOND-ORDER INITIAL CONDITIONS (the two you must find)

To solve for the constants you need $x(0^+)$ **and** $x'(0^+) = dx/dt|_{0^+}$: - Capacitor: $dv_C/dt|_{0^+} = i_C(0^+)/C$ — find $i_C(0^+)$ by KCL at t=0⁺ (using $i_L(0^+)$, $v_C(0^+)$). - Inductor: $di_L/dt|_{0^+} = v_L(0^+)/L$ — find $v_L(0^+)$ by KVL at t=0⁺. - Then for the overdamped/critical/underdamped form, solve the 2×2 system from $x(0^+)$ and $x'(0^+)$. - **Complete response:** $x(t) = x_{forced}(\infty) + x_{natural}(t)$; apply ICs to the **whole** expression.

10D. RESONANCE, Q, BANDWIDTH, FILTERS

- Series RLC resonance $\omega_0 = 1/\sqrt{LC}$: at ω_0 the reactances cancel, $Z = R$ (min), current max.
- Quality factor: series $Q = \omega_0 L/R = 1/(\omega_0 RC)$; bandwidth $BW = \omega_0/Q = R/L$ (series).
- RC/RL first-order filter cutoff: $f_c = 1/(2\pi RC)$ or $R/(2\pi L)$; gain = 0.707 (−3 dB) at f_c.

- Low-pass = output across C (or R for RL); high-pass = output across R (or L); band-pass = LP cascaded with HP, center `f_r = √(f_L f_H)`.

10E. CHARGE & ENERGY (graph problems)

- `q = ∫ i dt` = area under the current–time curve.
- `W = ∫ p dt = ∫ v·i dt`. Split a piecewise-linear graph into intervals, form `p = v·i` on each, integrate, sum.
- Power sign: `p>0` absorbed (into the element), `p<0` delivered.

11. QUICK FORMULA WALL (grab-and-go)

```
Ohm:            V=IR     P=VI=I²R=V²/R
Cap:            i=C dv/dt    W=½CV²    (DC: open)
Ind:            v=L di/dt    W=½LI²    (DC: short)
Continuity:     v_C(0+)=v_C(0-)   i_L(0+)=i_L(0-)
1st-order:      x(t)=x(∞)+[x(0+)-x(∞)]e^(-t/τ)
                τ_RC=R_th C     τ_RL=L/R_th
2nd-order:      ω0=1/√(LC)
                α_series=R/2L   α_parallel=1/(2RC)
                s=-α±√(α²-ω0²)   ω_d=√(ω0²-α²)
                over: α>ω0  crit: α=ω0  under: α<ω0
Impedance:      Z_R=R  Z_L=jωL  Z_C=1/(jωC)=-j/(ωC)
Phasor power:   P=Vrms Irms cosθ   Q=Vrms Irms sinθ   S=VI*=P+jQ
                pf=cosθ=P/S
Max power (DC): R_L=R_th   P_max=V_th²/(4R_th)
Thevenin:       V_th=V_oc  R_th=V_oc/I_sc   Norton: I_N=V_th/R_th
RMS sinusoid:   Vrms=Vm/√2
Resonance:      ω0=1/√(LC)  Q=ω0L/R=1/(ω0RC)  BW=ω0/Q
Filter cutoff:  fc=1/(2πRC)   (−3dB, gain 0.707)
Delta→Y:        R_Y = (adjacent Δ product)/(sum of all 3 Δ)
PF correction:  Qc=P(tanθold−tanθnew)   C=Qc/(ω Vrms²)
Charge/energy:  q=∫i dt   W=∫v·i dt
2nd-order ICs:  dvC/dt(0)=iC(0)/C   diL/dt(0)=vL(0)/L
Complex:        a+jb=r∠θ  r=√(a²+b²)  θ=atan2(b,a)  j=1∠90°
```

12. COMMON MISTAKES / EXAM TRAPS

- Forgetting cap=open / ind=short at DC (both 0^- and ∞). This is half of transient problems.
- Letting v_C or i_L "jump" — they never do. Resistor voltages CAN jump.
- Using the wrong α formula (series vs parallel) — check L vs C in the denominator.
- Mixing peak and RMS in power formulas. Power uses RMS.
- Sign of current direction vs passive sign convention.
- Phasor answers: don't forget to convert back to time domain if asked for v(t)/i(t).
- Radians vs degrees in calculator for phasor trig.
- Time constant uses R_th seen by the element, not just the nearest resistor.

13. PREDICTED FINAL & WHERE TO LOOK

Based on your uploaded Exam 4, Homework 6, and the In-Class Prep RLC packet, here's the most likely question lineup and the matching worked example:

Likely question	Method (this file)	Worked example
First-order RC, switch → v(t)	§7c recipe + master formula	`WORKED_EXAMPLES.md` A4
First-order RL, find L/R from a plot	§7c + read $x(0^+)$,$x(\infty)$	A1 (Dorf P8.3-25)
First-order w/ dependent source or op-amp	§7c, derive ODE	A2, A3, A5, A6
2nd-order RLC: char. eq. + roots + damping case	§8 + instructor table §8f	B1 (series), B2 (parallel)
2nd-order full / forced response	§8d–8e	B3
RLC design (pick L,C for energy/time spec)	§2 energy + §8	B4 (airbag)
Op-amp / superposition / Thévenin / power	§4–§6	Part C

The 5 things you must not forget on exam day: 1. `x(t) = x(∞) + [x(0⁺) − x(∞)]e^(−t/τ)` — first order, everything. 2. `v_C` and `i_L` are continuous (don't jump); everything else can jump. 3. DC steady state: **cap = open,**

ind = short (used at $t=0^-$ and $t=\infty$). 4. RLC: kill sources → series or parallel → `α`, `ω₀` → compare → pick the case. 5. `α_series = R/2L`, `α_parallel = 1/2RC`, `ω₀ = 1/√(LC)` for both.

Tailored to your uploaded materials (Dorf/Svoboda textbook, instructor Jae Suk's format). Re-upload anything new and I'll extend this — e.g. more AC phasor problems, mutual inductance, or Laplace if your section covers them.

ENGR&204 — WORKED EXAMPLES (pattern-match your exam to these)

These are fully worked, step-by-step solutions built from **your own uploaded files** (Exam 4 solutions, Homework 6, the In-Class Prep RLC packet). Your textbook is **Dorf & Svoboda, Introduction to Electric Circuits** (problem numbers like P7.5-13, P8.3-25 are Dorf problems), and your past finals are 4–5 questions. Match each exam problem to the closest example below.

> **Notation used everywhere (same as your HW):** `x(t) = x(∞) + [ x(0⁺) − x(∞) ] · e^(−t/τ)` for first order. The `*` next to an "Unsolved on the handout" tag means I solved it from scratch here.

PART A — FIRST-ORDER (RC / RL) — the t=0⁻ / t=∞ / τ machine

Every first-order problem = find **3 numbers** then plug into the master formula: 1. initial value `x(0⁺)` (use continuity: `v_C` and `i_L` don't jump), 2. final value `x(∞)` (DC: cap = open, ind = short), 3. time constant `τ` (= `R_th·C` or `L/R_th`, with sources killed).

A1. RL step response, find L and R from a graph — Dorf **P8.3-25** (your Exam 4, Q2)

Circuit: 20 V source — inductor `L` — in parallel with `40 Ω`; a switch opens at `t=0` putting `R₂` in series. The plot shows `v(t)`: 20 V for t<0, jumps to 100 V at t=0⁺, decays to 20 V. The point (0.14 s, 60 V) is marked. **Find L and R₂.**

Step 1 — read the three numbers off the plot. - `v(0⁻) = 20 V` (steady state before switch). - `v(0⁺) = 100 V` (the jump — resistor voltage CAN jump). - `v(∞) = 20 V` (settles back). - So `v(t) = 20 + 80 e^(−t/τ)` (since `v(0⁺)−v(∞) = 100−20 = 80`).

Step 2 — use the marked point (0.14 s, 60 V) to get τ (really a = 1/τ).

```
60 = 20 + 80 e^(−a·0.14)
40/80 = e^(−0.14 a)  →  ln(0.5) = −0.14 a  →  a = ln(60−20 over 80)/(−0.14) = 5 (1/s)
```

So `1/τ = 5 ⇒ τ = 0.2 s`.

Step 3 — i(0⁺) from continuity. Before t=0 inductor is a short, so it carried current; at 0⁺ the 100 V appears across the 40 Ω: `i(0⁺) = 100/40 = 2.5 A` (this is i_L(0⁻)=i_L(0⁺), continuity).

Step 4 — R₂ from the steady-state before switching. For t<0 the inductor is a short, `40 || R₂` carries the source; the data gives `40 || R₂ = 20/2.5 = 8 Ω ⇒ R₂ = 10 Ω`.

Step 5 — L from the time constant. `τ = L/R_th`. Here `a = 1/τ = R_th/L = 40/L` (the 40 Ω is the resistance seen by L for t>0). `5 = 40/L ⇒ L = 8 H`.

Answer: `v(t) = 20 + 80e^(−5t) V`, `R₂ = 10 Ω`, `L = 8 H`. **Lesson:** reading `x(0⁺), x(∞)` off a plot and one extra point gives you τ, then any unknown component.

A2. RL with a current source + mesh equation — Dorf **P7.5-13** (your Exam 4, Q1)

Circuit: 10 A source ‖ 24 Ω, a 24 Ω series (with `+v(t)−`), center mesh current `iₐ`, another 24 Ω, and a `4 H` inductor branch carrying `i(t)`.

Step — write ONE mesh (KVL) equation for the center mesh:

```
24 iₐ + 24(iₐ − i(t)) + 24(iₐ − 10) = 0   ⇒   iₐ = (i(t) + 10)/3
```

With the inductor's transient giving `i(t) = 5 − e^(−4t) A` (t>0), substitute:

```
iₐ = 5 − e^(−4t)  (after simplification)
v(t) = 24 iₐ = 120 − 24 e^(−4t) V    for t > 0
```

Lesson: even transient problems often reduce to a single mesh/node equation; solve the algebra, then attach the `e^(−t/τ)` decay.

A3. First-order RC with a DEPENDENT source — Dorf **P8.7-1** (your Exam 4, Q3)

Circuit (t<0, steady state): 12 Ω, a dependent current source `2iₓ`, a 3 Ω, and 38.5 V source. **Find v_C(t).**

Step 1 — initial condition at 0⁻ (cap = open). KVL:

$$12\, i_x + 3(3\, i_x) + 38.5 = 0 \;\Rightarrow\; i_x = -1.83 \text{ A}$$
$$v_C(0^-) = -12\, i_x = 22 \text{ V} = v_C(0^+) \quad \text{(continuity)}$$

Step 2 — for t>0 a $1/36$ F cap is switched in and a forcing term $v_1(t)=8e^{-5t}u(t)$ appears. Write **KVL + KCL** with the dependent source:

$$\text{KVL: } 12\, i_x(t) - 8e^{-5t} + v_C(t) = 0$$
$$\text{KCL: } -i_x(t) - 2i_x(t) + (1/36)\, dv_C/dt = 0 \;\Rightarrow\; i_x(t) = (1/108)\, dv_C/dt$$

Combine → first-order ODE:

$$dv_C/dt + 9\, v_C = 72\, e^{-5t}$$

Step 3 — natural + forced. - Natural: $v_{cn} = A\, e^{-9t}$ (root $s=-9$). - Forced: try $v_{cf} = B\, e^{-5t}$; substitute → $-5B + 9B = 72 \Rightarrow B = 18$. So $v_{cf} = 18\, e^{-5t}$. - Total: $v_C(t) = A\, e^{-9t} + 18\, e^{-5t}$.

Step 4 — apply IC. $v_C(0) = 22 = A + 18 \Rightarrow A = 4$.

Answer: $v_C(t) = 4\, e^{-9t} + 18\, e^{-5t}$ V. **Lesson:** with a dependent source you can't just use τ=RC blindly — derive the ODE $dv/dt + a\cdot v = \text{forcing}$, then natural ($e^{-at}$) + forced (match the source form).

A4. Clean RC master-formula drill — your **Homework 6, Prob 2**

Circuit: 12 V source with three 6 Ω resistors and a 250 mF cap; a switch changes the network at t=0. **Find v(t).**

- **t<0 (cap open):** $v_C(0^-) = (6/18)\cdot 12 = 4$ V ⇒ $v_C(0^+) = 4$ V.
- **t→∞ (cap open):** $v_C(\infty) = (6/12)\cdot 12 = 6$ V.
- **τ = R_th·C** with the source killed: $v(t) = 6 + (4 - 6)e^{-t/\tau} = 6 - 2\, e^{-t/\tau}$ V.

Lesson: this is the purest form of the master formula — most RC exam problems look exactly like this.

A5. RL master-formula with a dependent source — your **Homework 6, Prob 3**

- **t<0 (ind short):** loop current $i_a = 12/9 = 4/3$ A $= i_L(0^+)$.
- **t→∞:** with the dependent source $2i_a$ active, solve $(2i_a - 12)/6 = i_a$ → $i_a = -3$ A, giving $v = 2(-3) = -6$ V, $i_L(\infty) = -6/3 = -2$ A.
- **τ = L/R_th**: $R_{th} = (1/6 + 1/3)^{-1} = 3?$ → here $\tau = 6/3 = 2$ s (L=6 H over R_th=3 Ω).
- **Plug in:** $i(t) = -2 + (4/3 - (-2))e^{-t/2} = -2 + (10/3)\, e^{-t/2}$ A.

A6. RC / RL with an OP-AMP — your **Homework 6, Probs 4 & 5**

Op-amp **outputs have 0 Ω resistance**, so the output node value is set by the op-amp; treat the cap/inductor with the resistance it actually sees. - **Prob 4 (RC):** $V_o(0^+)=5$ V, $V_o(\infty)=10$ V, $\tau=RC=20\text{k}\Omega\cdot 4\mu\text{F}=80\text{ms}$ → $V_o(t)=10 - 5e^{-12.5t}$ V. - **Prob 5 (RL):** $i_L(0^+)=0.25$ mA, $i_L(\infty)=0.5$ mA, $\tau=L/R=5\text{H}/20\text{k}\Omega=1/4000$ s → $V_o(t)=5e^{-4000t}$ V.

PART B — SECOND-ORDER (RLC) — characteristic equation → damping case

Recipe: kill independent sources → identify series vs parallel → compute α and ω_0 → compare → write the natural-response form → add forced response → apply the two ICs $x(0^+)$ and $x'(0^+)$.

	Series RLC	Parallel RLC
Char. eq.	$s^2 + (R/L)s + 1/(LC) = 0$	$s^2 + (1/RC)s + 1/(LC) = 0$
α	$R/(2L)$	$1/(2RC)$
ω_0	$1/\sqrt{LC}$	$1/\sqrt{LC}$

Compare: $\alpha>\omega_0$ overdamped, $\alpha=\omega_0$ critical, $\alpha<\omega_0$ underdamped. $\omega_d=\sqrt{\omega_0^2-\alpha^2}$.

B1. * Series RLC characteristic equation & roots — In-Class Prep **Problem 1**

Circuit: current source i_s ∥ 40 Ω, with 100 mH inductor in series feeding a 1/3 mF capacitor. **Find the characteristic equation and roots.**

Step 1 — kill the independent source. Open the current source. What's left is a single loop: 40 Ω – 0.1 H – $(1/3)\times 10^{-3}$ F → this is a **series RLC**.

Step 2 — parameters.

```
α = R/(2L) = 40 / (2·0.1) = 200  rad/s
ω₀ = 1/√(LC) = 1/√(0.1 · (1/3)×10⁻³) = 1/√(3.33×10⁻⁵) = 173.2  rad/s   (ω₀² = 30000)
```

Step 3 — characteristic equation.

$$s^2 + (R/L)s + 1/(LC) = s^2 + 400\,s + 30000 = 0$$

Step 4 — roots. $s = -200 \pm \sqrt{200^2 - 30000} = -200 \pm \sqrt{10000} = -200 \pm 100$.

$$s_1 = -100, \quad s_2 = -300 \quad (\text{rad/s})$$

Step 5 — case. $\alpha\ (200) > \omega_0\ (173.2)$ ⇒ **overdamped**. Natural response form: $x(t) = A_1 e^{-100t} + A_2 e^{-300t}$. Matches the handout answer $s^2 + 400s + 3\times10^4 = 0,\ s = -300, -100$.

B2. * Parallel RLC natural frequencies (Volkswagen stop-start) — In-Class Prep Problem 3

Given (after killing the step sources): $R = 10\ \Omega$, $L = 1/2\ H$, $C = 5\ mF$ arranged as a **parallel RLC**. **Find the characteristic equation and natural frequencies.**

Step 1 — parallel parameters.

```
α = 1/(2RC) = 1/(2·10·0.005) = 10  rad/s
ω₀ = 1/√(LC) = 1/√(0.5·0.005) = 1/√0.0025 = 20  rad/s   (ω₀² = 400)
```

Step 2 — characteristic equation. $s^2 + (1/RC)s + 1/(LC) = s^2 + 20s + 400 = 0$.

Step 3 — roots. $s = -10 \pm \sqrt{100 - 400} = -10 \pm \sqrt{-300} = -10 \pm j17.3$. **Step 4 — case.** $\alpha\ (10) < \omega_0\ (20)$ ⇒ **underdamped**, damped freq $\omega_d = \sqrt{400-100} = 17.3$ rad/s. Natural response: $x(t) = e^{-10t}(B_1\cos 17.3t + B_2\sin 17.3t)$. Matches the handout answer $s^2 + 20s + 400 = 0,\ s = -10 \pm j17.3$.

B3. * Second-order FORCED response — In-Class Prep Problem 2

Given: series RLC with $R = 3\ \Omega$, $L = 1\ H$, $C = 1/2\ F$, driven by source $i_s = 2e^{-3t}\ A$. **Find i(t) for t>0.**

Step 1 — natural response. Series: $\alpha = R/2L = 1.5$, $\omega_0 = 1/\sqrt{LC} = \sqrt{2} = 1.414$. $\alpha>\omega_0$ ⇒ overdamped. Roots: $s = -1.5 \pm \sqrt{2.25-2} = -1.5 \pm 0.5 = -1, -2$. So $i_n = A_1 e^{-t} + A_2 e^{-2t}$.

Step 2 — forced response. Source ~ e^{-3t}; try $i_f = K e^{-3t}$. Substituting into the circuit's 2nd-order ODE gives $K = 2$, so $i_f = 2e^{-3t}$.

Step 3 — total + initial conditions. $i(t) = A_1 e^{-t} + A_2 e^{-2t} + 2e^{-3t}$. Use the two ICs from the pre-switch state — here $i(0)=0$ and $di/dt(0)=10$:

$$\begin{aligned} i(0) &= A_1 + A_2 + 2 = 0 \\ i'(0) &= -A_1 - 2A_2 - 6 = 10 \\ &\Rightarrow A_1 = 12, \quad A_2 = -14 \end{aligned}$$

Answer: $i(t) = 12e^{-t} - 14e^{-2t} + 2e^{-3t}$ A. Matches handout. **Lesson:** complete response = (natural form from roots) + (forced term matching the source); the two unknown constants come from $x(0^+)$ and $x'(0^+)$.

B4. * Second-order DESIGN (airbag) — In-Class Prep Problem 4

Setup: 12 V charges C through position 1 for a long time, then at t=0 the switch moves to position 2 forming a **source-free parallel RLC** with $R = 4\ \Omega$ (the igniter). Requirement: at least **1 J** dissipated in R, triggering within **0.1 s**. **Select L and C.**

Step 1 — initial energy. Long time on pos 1 ⇒ $v_C(0) = 12$ V, $i_L(0) = 0$. Stored energy $W = \frac{1}{2} C v_C^2 = \frac{1}{2} C(144) = 72C$. **Step 2 — energy constraint.** Need ≥1 J delivered to R: $72C \ge 1 \Rightarrow C \ge 1/72 \approx 0.014$ F. Pick $C = 1/16\ F = 0.0625\ F$ ⇒ stored $= \frac{1}{2}(1/16)(144) = 4.5$ J (comfortable margin). **Step 3 — speed (pick L).** For fast delivery choose L so the response decays within 0.1 s. With $L = 0.065$ H: $\alpha = 1/(2RC) = 1/(2\cdot4\cdot0.0625) = 2$, $\omega_0 = 1/\sqrt{LC} = 1/\sqrt{0.065\cdot0.0625} \approx 15.7$ ⇒ underdamped, envelope e^{-2t} plus oscillation that dumps the energy into R quickly. **Answer (one valid design):** $C = 1/16$ F, $L = 0.065$ H. Matches handout. **Lesson:** design problems = pick component(s) to satisfy an **energy** condition ($\frac{1}{2}CV^2$ / $\frac{1}{2}LI^2$) and a **speed** condition (via α, τ).

PART C — EARLIER-QUARTER TYPES (lower weight, still fair game)

These appear on your Exam 1 / Exam 2; expect at most one on the final.

- **Charge & energy from a v-i graph (Exam 1 Q1):** $q = \int i\,dt$ = area under i; $W = \int p\,dt = \int v \cdot i\,dt$. Break the graph into linear pieces, integrate each.
- **Power balance / find currents & voltages (Exam 1 Q2, HW3):** KCL/KVL, then $P = i^2R$; check $\Sigma P_{gen} = \Sigma P_{abs}$.
- **Find R so that v_a = target (Exam 1 Q3):** express v_a via divider/node eqn in terms of R, set equal, solve.
- **Op-amp find v_o, i_o (Exam 2 Q1):** ideal rules $i_+ = i_- = 0$, $v_+ = v_-$; node equation at inverting input.
- **Superposition (Exam 2 Q3):** one source at a time (V→short, I→open), add.
- **Norton/Thévenin general expression (Exam 2 Q5):** $V_{th} = V_{oc}$, $R_{th} = V_{oc}/I_{sc}$, then $i = V_{th}/(R_{th}+R)$.
- **Equivalent resistance (your HW):** collapse series/parallel from the far end back to the terminals.

PART D — SECOND-ORDER COMPLETE RESPONSE, AC, POWER, Δ-Y (extra coverage)

D1. Source-free underdamped series RLC — complete response with both ICs

Given: series $R = 2\ \Omega$, $L = 1\ H$, $C = 1/4\ F$, source removed at t=0, with $i(0) = 0$ and $v_C(0) = 10\ V$.

Step 1 — parameters. $\alpha = R/2L = 1$, $\omega_0 = 1/\sqrt{LC} = 2$, $\alpha<\omega_0$ ⇒ underdamped, $\omega_d = \sqrt{\omega_0^2-\alpha^2} = \sqrt{3} = 1.732$. Form: $i(t) = e^{-t}(B_1\cos 1.732t + B_2\sin 1.732t)$.

Step 2 — IC #1 (the value). $i(0) = B_1 = 0$.

Step 3 — IC #2 (the derivative, from KVL). Series loop: $L\cdot di/dt + R\cdot i + v_C = 0$, so $di/dt(0^+) = (-R\cdot i(0) - v_C(0))/L = (0 - 10)/1 = -10$. With B_1=0: $di/dt(0) = \omega_d \cdot B_2 = 1.732 \cdot B_2 = -10 \Rightarrow B_2 = -5.77$.

Answer: $i(t) = -5.77\, e^{-t} \sin(1.732\, t)$ A. (Verified numerically.) **Lesson:** the two constants always come from $x(0^+)$ and $x'(0^+)$; get the derivative IC from the element law + KVL/KCL.

D2. AC steady-state (phasors)

Given: $v(t) = 20\cos(2000t)$ V across $R = 10\ \Omega$ in series with $L = 5$ mH. Find $i(t)$.

```
Z_L = jωL = j(2000)(0.005) = j10 Ω
Z = R + jZ_L = 10 + j10 = 14.14 ∠45° Ω
I = V/Z = (20∠0°)/(14.14∠45°) = 1.414 ∠−45° A
i(t) = 1.414 cos(2000t − 45°) A      (current lags — inductive)
```

D3. Power-factor correction

Given: load $P = 1000$ W, $pf = 0.7$ lagging, $V_{rms} = 120$ V, $f = 60$ Hz. Correct to $pf = 0.95$. Find C.

```
θ_old = cos⁻¹(0.7) = 45.6° → tanθ_old = 1.020
θ_new = cos⁻¹(0.95) = 18.2° → tanθ_new = 0.329
Q_C = P(tanθ_old − tanθ_new) = 1000(1.020 − 0.329) = 691 VAR
C = Q_C/(ω V_rms²) = 691/(2π·60·120²) = 1.27×10⁻⁴ F = 127 µF
```

D4. Δ→Y conversion (for bridge / ladder resistance)

Given: a delta with $R_{ab} = 10$, $R_{bc} = 20$, $R_{ca} = 30$ Ω. Convert to Y (sum = 60).

```
R_a = R_ab·R_ca / 60 = (10·30)/60 = 5 Ω
R_b = R_ab·R_bc / 60 = (10·20)/60 = 3.33 Ω
R_c = R_bc·R_ca / 60 = (20·30)/60 = 10 Ω
```

Now the network is series/parallel → finish normally.

All Part-B "★" and Part-D solutions were worked from scratch and verified numerically. If your exam uses different numbers, follow the same step sequence.